The $qx + 1$ Generalization of Tao's Syracuse Measure: A Closed-Form Pressure Equation, an Endogeny Barrier, and Its Convergent Characterization Across Five Independent Vocabularies

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Abstract

We study a natural $qx+1$ generalization of Tao's Syracuse random variable [1], the probabilistic object underlying the strongest unconditional partial result toward the Collatz conjecture. We prove an exact identity, $\rho_{\text{ann}}(\alpha) = q^{\alpha-1}/(2^\alpha - 1)$, for the *annealed* pressure of the natural multiplicative population functional on the reverse tree of the accelerated $qx + 1$ map — the exponential growth rate, averaged over all root residues at a given depth, of the associated partition function — via an exact fibre-counting bijection, correcting a finite-automaton argument for the same identity that we show is not well posed. The nontrivial root of $\rho_{\text{ann}}(\alpha) = 1$ undergoes a structural transition at $q = 5$: it equals exactly $\alpha^* = 2$ at $q = 3$ and falls strictly below 1 for every $q \geq 5$, conjecturally signalling a qualitative change from positive to zero natural density of the reverse tree (for $q = 3$, exactly the classical Growth Exponent Conjecture of Applegate–Lagarias). We show, moreover, that the two roots of this equation always sit on opposite sides of a quenched/annealed freezing transition in the sense of directed polymers in a random environment: the smaller root is always unfrozen, so the annealed identity transfers rigorously to the quenched counting function of the matching i.i.d. branching-random-walk model, and, we conjecture on strong empirical grounds, to the genuine arithmetic tree itself; the larger root is always frozen, so it does *not* by itself establish the tail index of the martingale scale factor governing the sub-exponential correction term, a fact we verify independently by exact computation. At $q = 3$ this larger root, $\alpha^* = 2$, still matches the tail exponent established empirically (independently of the pressure argument) by three separate methods in earlier work on the classical Collatz map; for $q \geq 5$ the analogous claim is left as an explicit, weakly-supported conjecture. We then identify and formally characterize an *endogeny barrier*, in the technical sense of Aldous–Bandyopadhyay [7], obstructing any proof that relies only on the linear recursive structure of the tree: solutions of the form $G = W \cdot Y$, with Y an arbitrary bounded factor invariant along the tree, solve the same fixed-point recursion as the canonical ("endogenous") solution W and are statistically indistinguishable from it by every observable we are able to test. We show, by four methodologically independent vocabularies (endogeny itself counting as the first), that closing this gap is equivalent to one and the same unresolved arithmetic statement: Wirsching's 1998 Weak Covering Conjecture [4] and Tao's 2020 $\beta = 1$ conjecture [3] (which we show is literally the same algebraic object, hence counted once), a functional equation for an explicit base-3 analogue of the Fabius function arising in Wirsching's 2003 paper [5], and two candidate lemmas suggested as natural next steps by Tao's bivariate Fourier-decay technique [1, 2] (also counted once, since both share the same second-moment Fourier decomposition), both of which we execute to completion and both of which fail through the *same* missing ingredient: one yields a genuine closed-form structure theorem together with a conditional reduction whose hypothesis (square-integrability of the Syracuse density on $\mathbb{Z}_3$) we refute by an exact recursive computation; the other yields a genuine but strictly weaker conditional dichotomy together with an exact negative result, via Jensen's formula, showing that the relevant Littlewood–Offord/Riesz-product machinery has an intrinsic ℓ^1 Fourier-budget slack of a factor ≈ 1.88 in the regime that matters. We complement this theoretical package with a fully independent, purely empirical result: implementing the exact reverse tree of the $5x+1$ map and applying Aitken Δ^2 extrapolation to a sequence of nested-window measurements, we resolve a standing dispute between two competing predictions for the $5x+1$ counting exponent (Kontorovich–Lagarias [6] versus a competing stochastic model due to Volkov, both discussed in [6]), measuring 0.639 (95% CI $[0.633, 0.645]$) and decisively excluding the competing value 0.678. Finally, a directed literature search situates the barrier precisely within the wider theory of Fourier decay for self-similar and self-conformal measures [25, 26], and shows it is structurally inaccessible to that toolkit because of the rigidity of closed subgroups of $\mathbb{Z}_3^\times$. We argue that this convergence of five methodologically independent and technically unrelated vocabularies onto the same unproven statement — two of whose pairwise coincidences are themselves proved identities, not analogies, and are counted once each — is itself significant evidence that the obstruction is a genuine feature of the problem rather than an artifact of any single method, and we state precisely what remains to be proved.

Contents

1 Introduction

1.1 The Collatz conjecture and the Syracuse random variable

The Collatz (or $3x + 1$) conjecture asserts that iterating the map

$$C(n) = \begin{cases} n/2 & n \text{ even} \\ 3n+1 & n \text{ odd} \end{cases}$$

started at any positive integer n eventually reaches 1. Despite its elementary statement, the conjecture remains open, and the strongest unconditional result toward it is due to Tao [1], who showed that *almost all* orbits (in the sense of logarithmic density) attain almost bounded values. Tao's proof introduces the *Syracuse random variable*: fixing n, let $a_1, a_2, \ldots, a_n$ be i.i.d. geometric random variables with $\Pr(a_i = k) = 2^{-k}$ for $k \geq 1$, and define

$$F_n(a) := \sum_{j=1}^{n} 3^{j-1}\, 2^{-a_{[1,j]}} \pmod{3^n}, \qquad a_{[1,j]} := a_1 + \cdots + a_j, \tag{1}$$

where 2^{-1} denotes the multiplicative inverse of 2 modulo 3^n. The random variable $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) := F_n(a_1, \ldots, a_n)$ models, in a precise sense, the residue of an accelerated Collatz orbit modulo 3^n after n odd steps, and Tao's proof reduces the conjecture's target statement to controlling the distribution of this random variable, in particular via a Fourier decay estimate:

Proposition 1.1 (Tao 2022, Prop. 1.17). *For every $A > 0$ there is C_A such that for all $n \geq 1$ and every $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ with $3 \nmid \xi$,*

$$\left|\mathbb{E}\, e\big(\xi\, F_n(a)/3^n\big)\right| \leq C_A\, n^{-A},$$

with C_A uniform in n and ξ (subject only to $3 \nmid \xi$).

This is a super-polynomial (though not exponential) decay estimate, and it is the technical heart of Tao's theorem. A recursive consequence of the construction (1), which we use repeatedly below, is the following *exact self-similarity* identity, verified directly against Tao's text:

Proposition 1.2 (Tao 2022, eq. (1.23)). *For $k \leq n$,*

$$\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) \bmod 3^k \;\equiv\; \mathrm{Syrac}(\mathbb{Z}/3^k\mathbb{Z})$$

as an identity in law. Equivalently, writing $F_n(a) = 2^{-a_1}\big(1 + 3\,F_{n-1}(a_2, \ldots, a_n)\big) \bmod 3^n$, the coarsest k coordinates of $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ *depend only on $(a_1, \ldots, a_k)$, the digits* adjacent to the root *in the indexing of (1), not on the deepest ones.*

The orientation convention embedded in Proposition 1.2 (that a_1, not a_n, is the coordinate adjacent to the root) is a recurring source of error when translating Tao's construction into reverse-tree language, and we flag it explicitly throughout.

1.2 This paper's contributions

Building on [1] and on the classical predecessor-density program of Wirsching [4, 5], we make the following contributions.

(i) **A closed-form pressure equation for $qx + 1$ (§3).** We show that the tail exponent controlling the reverse-tree population martingale of the accelerated $qx + 1$ map satisfies the exact equation $q^{\alpha-1} = 2^\alpha - 1$, with a structural transition at $q = 5$.

(ii) **A precise endogeny barrier (§4).** We exhibit an explicit family of *non-endogenous* solutions to the tree's fixed-point recursion, prove they are statistically indistinguishable from the canonical one by pressure, tail index, or linear regression against the limiting measure, and identify exactly which extra hypothesis (near-independence of fresh 3-adic digits across sibling subtrees) is needed to rule them out.

(iii) **An empirical bound on the residual coupling (§5).** A three-arm control experiment gives $\rho_{\text{eff}} \lesssim 0.06$ (95% CI) for the effective arithmetic coupling between sibling subtrees, converting the theoretical gap of (ii) into a quantitative empirical bound.

(iv) **An unconditional empirical result (§6).** We resolve, using exact enumeration and Aitken Δ^2 extrapolation, a long-standing dispute between two competing predictions for the $5x + 1$ counting exponent.

(v) **A direct computational test of the Weak Covering Conjecture (§7).**

(vi) **Three convergent regimes of precision for Tao's bivariate extension (§8), and triangulation with two further independent formulations of the same obstruction (§9).**

(vii) **Two candidate lemmas, executed to completion (§10).** Both fail through the same missing ingredient, yielding along the way a genuine structure theorem, a computational refutation of a natural sufficient condition, and an exact negative result via Jensen's formula.

(viii) **A literature-theoretic contextualization (§11).** We show the barrier is structurally inaccessible to the general theory of Fourier decay for self-similar measures, for a precise arithmetic reason (rigidity of closed subgroups of $\mathbb{Z}_3^\times$), and we identify a fifth, purely combinatorial, independent vocabulary for the same obstruction in very recent work [31].

We deliberately frame this paper not as an attempted proof of any part of the Collatz conjecture, but as a precise characterization of an obstruction that recurs, in essentially the same form, across every technique we or the existing literature have brought to bear on this specific line of attack. We return to this framing choice in §12.

2 Setup: the accelerated $qx + 1$ map and its reverse tree

Fix an odd integer $q \geq 3$ coprime to 2. The *accelerated $qx + 1$ map* is

$$T_q(n) = \frac{qn + 1}{2^{v_2(qn+1)}}, \qquad n \text{ odd},$$

where v_2 is the 2-adic valuation. For $q = 3$ this is the standard accelerated Collatz map. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node u, adjoining a child $w_a := (2^a u - 1)/q$ for every $a \geq 1$ such that $2^a u \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer, since an odd numerator divided evenly by an odd denominator yields an odd quotient. Writing $d := \mathrm{ord}_q(2)$ for the multiplicative order of 2 modulo q, admissibility of a for a given parent u depends only on $u \bmod q$: *if* some $a_0(u) \in \{1, \ldots, d\}$ satisfies $2^{a_0(u)} u \equiv 1 \pmod{q}$, it is unique, and the admissible exponents are exactly $a \equiv a_0(u) \pmod{d}$. Such $a_0(u)$ exists exactly when $u \bmod q$ lies in the subgroup $\langle 2 \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$ generated by 2; nodes with $u \equiv 0 \pmod{q}$, and, more generally, nodes whose residue mod q falls outside $\langle 2 \rangle$, are *sterile* (no admissible children). When 2 is a primitive root mod q (as for $q = 3, 5, 9$, all used numerically below), $\langle 2 \rangle$ is all of $(\mathbb{Z}/q\mathbb{Z})^\times$ and $u \equiv 0 \pmod{q}$ is the only sterile class; for $q = 3$ this recovers the familiar fact that multiples of 3 have trivial reverse subtrees. In general, however, $d = \mathrm{ord}_q(2)$ can be a proper divisor of $\varphi(q)$: at $q = 7$, $d = 3 < \varphi(7) = 6$ and $\langle 2 \rangle = \{1, 2, 4\}$, so $u \equiv 3, 5, 6 \pmod{7}$ are sterile too, not merely $u \equiv 0$. This does not affect the pressure identity of Theorem 3.3 below (whose proof, via Lemma 3.2, works in the reverse direction — from an unconstrained exponent sequence to its unique root — and never invokes existence of $a_0(u)$ for every u), nor the numerical enumeration of §3.4, whose implementation already excludes exactly these classes; it is a correction to this structural description only.

On this tree we study the linear population functional G satisfying the fixed-point recursion

$$G(u) = \sum_{a \equiv a_0(u)\ (d)} q \cdot 2^{-a}\, G(w_a), \tag{2}$$

which for $q = 3$ is exactly the martingale-type recursion underlying the $G(v)/\mu$ line of investigation summarized in §4 below, and which is the direct reverse-tree analogue of (1).

3 The closed-form pressure equation

3.1 The pressure operator: an exact annealed identity

A first, natural attempt to organize the recursion (2) into a transfer operator is to track the parent's residue class modulo q alone. This attempt fails, and it is worth recording exactly why, since an earlier draft of this paper asserted a version of it that does not survive scrutiny.

Example 3.1 (The naive mod-q automaton is not well defined). Take $q = 3$ (so $d = \mathrm{ord}_3(2) = 2$) and $a = 2$, admissible for every parent $u \equiv 1 \pmod 3$. Two parents $u = 1$ and $u = 7$, both $\equiv 1 \pmod 3$, give $w_2(1) = (4 - 1)/3 = 1$ and $w_2(7) = (28 - 1)/3 = 9$, with $w_2(1) \equiv 1 \pmod 3$ but $w_2(7) \equiv 0 \pmod 3$: the child's residue class depends on $u \bmod 9$, not merely on $u \bmod 3$. In general, writing $u = u_0 + qk$ with $u_0 = u \bmod q$, one finds $w_a \bmod q = [(2^a u_0 - 1)/q \bmod q] + [2^a k \bmod q]$: an explicit dependence on $k = \lfloor u/q \rfloor \bmod q$. No finite modulus q^k repairs this: the same computation shows $w_a \bmod q^k$ depends on $u \bmod q^{k+1}$, one digit further than what mod-q^k knowledge of u supplies. There is therefore no well-defined finite-state transfer operator tracking a single generation of the recursion on residue classes of any fixed finite modulus.

What *is* well defined, and is the correct replacement for the naive picture of Example 3.1, is a bijection between an entire admissible fibre at one level of the digit tower and the full residue ring one level below.

Lemma 3.2 (Fibre bijection)**.** *Fix $a \geq 1$ and $j \geq 1$. The map $v \mapsto w_a(v) := (2^a v - 1)/q$ is a bijection from $\{v \bmod q^j : 2^a v \equiv 1 \pmod q\}$ (an arithmetic progression of q^{j-1} residues mod q^j) onto $\mathbb{Z}/q^{j-1}\mathbb{Z}$.*

Proof. Well defined: $v \bmod q^j$ determines $2^a v - 1 \bmod q^j$, hence $w_a(v) \bmod q^{j-1}$. Injective: if $v \not\equiv v' \pmod{q^j}$ within the progression, write $v - v' = qt$ with $t \not\equiv 0 \pmod{q^{j-1}}$; then $w_a(v) - w_a(v') = 2^a t \not\equiv 0 \pmod{q^{j-1}}$ since 2 is a unit mod q^{j-1}. Surjectivity follows since the map is an injection between two finite sets of equal size q^{j-1}. □

Iterating Lemma 3.2 along a depth-k path $(a_1, \ldots, a_k)$ recovers, in truncated form, exactly the self-similar identity (10) of §11 below: every sequence $(a_1, \ldots, a_k) \in (\mathbb{Z}_{\geq 1})^k$, with no admissibility restriction on the individual a_i, is admissible for *exactly one* root residue $u_0 \bmod q^k$, namely $u_0 \equiv \sum_{i=1}^{k} q^{i-1} 2^{-S_i} \pmod{q^k}$, $S_i := a_1 + \cdots + a_i$. Define the depth-$k$ partition function rooted at u_0,

$$Z_k(\alpha; u_0) := \sum_{\substack{(a_1,\ldots,a_k)\text{ admissible} \\ \text{from } u_0}} \prod_{i=1}^{k} \left(q \cdot 2^{-a_i}\right)^{\alpha}.$$

Theorem 3.3 (Exact annealed pressure identity)**.** *For every odd $q \geq 3$, every $k \geq 1$, and every $\alpha > 0$,*

$$\sum_{u_0 \in \mathbb{Z}/q^k\mathbb{Z}} Z_k(\alpha; u_0) \;=\; \left(\frac{q^{\alpha}}{2^{\alpha} - 1}\right)^{k}, \tag{3}$$

so the average *of $Z_k(\alpha; \cdot)$ over all q^k root residues equals exactly $\left(q^{\alpha-1}/(2^{\alpha} - 1)\right)^k$; the associated annealed pressure is*

$$\rho_{\mathrm{ann}}(\alpha) = \frac{q^{\alpha-1}}{2^{\alpha} - 1}, \tag{4}$$

giving the pressure equation at $\rho_{\mathrm{ann}} = 1$:

$$q^{\alpha-1} = 2^{\alpha} - 1. \tag{5}$$

Proof. By Lemma 3.2 (iterated), the admissible depth-k sequences rooted at the q^k distinct residues u_0 partition the full, unconstrained set $(\mathbb{Z}_{\geq 1})^k$ with no overlap and no omission. Summing $\prod_i (q2^{-a_i})^\alpha$ over *all* sequences in $(\mathbb{Z}_{\geq 1})^k$ therefore factors as $\left(\sum_{a=1}^\infty (q2^{-a})^\alpha\right)^k = \left(q^\alpha/(2^\alpha - 1)\right)^k$, which is exactly the left-hand side of (3) summed over the partition. □

We verified (3) independently by direct enumeration for $q \in \{3, 5, 7, 9\}$, $k \in \{1, 2, 3\}$, and $\alpha \in \{0.26, 0.5, 1, 1.37, 2\}$, matching to the truncation error of the (rapidly convergent) tail of the sum over a.

Remark 3.4 (Mass conservation as a corollary). The right-hand side of (3) evaluates to exactly $1^k = 1$ at $\alpha = 1$ for every q (mass conservation of the branching process). Thus $\alpha = 1$ is always a root of (5).

3.2 Quenched vs. annealed: a freezing transition

Theorem 3.3 is a statement about the *average* of $Z_k(\alpha; \cdot)$ over root residues; it is not, by itself, a statement about $Z_k(\alpha; u_0)^{1/k}$ for a single, fixed root u_0 — which is what actually governs the population martingale $G(v)$ for one specific integer v. Whether the two coincide is exactly the quenched-versus-annealed question familiar from directed polymers in a random environment and from Derrida's random energy model, governed by a standard freezing criterion. Writing $P(\alpha) := \log \rho_{\text{ann}}(\alpha)$ for the annealed log-pressure and $s(\alpha) := P(\alpha) - \alpha P'(\alpha)$ for the associated entropy, quenched and annealed rates coincide where $s(\alpha) > 0$ (an *unfrozen* phase) and diverge where $s(\alpha) < 0$ (a *frozen* phase, in which the quenched rate is dominated by rare, low-entropy paths rather than by the bulk of the ensemble).

Proposition 3.5 (The larger root is always frozen)**.** *For every odd $q \geq 3$, writing $\alpha_- < \alpha_+$ for the two positive roots of $\rho_{\text{ann}}(\alpha) = 1$: $s(\alpha_-) > 0$ (unfrozen) and $s(\alpha_+) < 0$ (frozen), unconditionally.*

Proof. At any root, $P(\alpha) = 0$, so $s(\alpha) = -\alpha P'(\alpha)$. Since $\rho_{\text{ann}} = e^P$ is log-convex (§4.4), P itself is strictly convex; with exactly two zeros $\alpha_- < \alpha_+$, necessarily $P'(\alpha_-) < 0 < P'(\alpha_+)$ (the function must be decreasing through the first zero and increasing through the second, on either side of its unique minimum). Hence $s(\alpha_-) = -\alpha_- P'(\alpha_-) > 0$ and $s(\alpha_+) = -\alpha_+ P'(\alpha_+) < 0$. □

For $q \geq 5$, where $\alpha_+ = 1$ always, this has an exact closed form: at $\alpha = 1$, $2^\alpha - 1 = 1$ so $\log(2^\alpha - 1) = 0$, and $s(1) = 2 \log 2 - \log q = \log(4/q)$, which is negative exactly when $q > 4$. So the freezing of the trivial root for every odd $q \geq 5$ is not a numerical coincidence but the same sign condition as the mean-drift remark above ($\log q - 2 \log 2 > 0$): the drift that makes orbits expand on average is, up to a minus sign, exactly the entropy deficit that freezes $\alpha_+ = 1$.

We computed the freezing threshold $\alpha_c(q)$ (the root of $s(\alpha) = 0$, which by Proposition 3.5 always lies strictly between $\alpha_-(q)$ and $\alpha_+(q)$) directly:

q	$\alpha_c(q)$
3	1.355
5	0.794
7	0.564
9	0.437

confirming $\alpha_-(q) < \alpha_c(q) < \alpha_+(q)$ in every case, as Proposition 3.5 requires. We also verified numerically, with an independent memoised recursion computing $Z_k(\alpha; u_0)$ exactly at a fixed (not averaged) root, that the freezing is not merely a formal entropy computation but a genuine divergence of quenched from annealed growth: at $q = 3$, $u_0 = 1$ — a genuine period-one cycle of the residue dynamics, since $w_2(1) = 1$ exactly — $Z_k(2;1)^{1/k}$ shows an oscillation with geometric-mean ratio well below the annealed rate $\rho_{\text{ann}}(2) = 1$; at $q = 5$, $Z_k(1;u_0)^{1/k}$ at the roots $u_0 \in \{1,2,3,4\}$ likewise fails to track $\rho_{\text{ann}}(1) = 1$. We flag, rather than obscure, a limitation of this specific check: every one of these roots lies on or feeds directly into the smallest-admissible-branch map's own trivial cycle ($1 \leftrightarrow 3$ for $q = 5$, exactly as $u_0 = 1$ does for $q = 3$), so the divergence observed here has a deterministic-recurrence explanation available in addition to the entropy mechanism of Proposition 3.5, and does not by itself sample the generic (non-periodic) environment that the quenched statement is really about. We therefore report this as an illustration of freezing at a special, easily-computed class of roots, not as a confirmation at a generic one; §3.4 below and the statistically calibrated measurement of §6 — both averaging over many roots sampled well away from any known short cycle — carry the actual generic-environment evidence.

The consequence, by Proposition 3.5, is unconditional: Theorem 3.3, though exact, rigorously transfers only to the *counting-exponent* half of Theorem 3.6 below (the smaller root $\alpha_-(q)$, always unfrozen); it never establishes, for any q, the tail index of the martingale scale factor W_u also claimed there, since the larger root $\alpha_+(q)$ governing that claim is always frozen. We treat this honestly as Conjecture 3.10 below, rather than as a further consequence of Theorem 3.3.

3.3 The structural transition at $q = 5$

Besides the trivial root $\alpha = 1$, equation (5) has exactly one further positive root $\alpha^*(q)$ (by strict log-convexity of $\alpha \mapsto q^{\alpha-1}/(2^\alpha - 1)$ on $\alpha > 0$, discussed in §4.4 below, the equation $\rho(M_q(\alpha)) = 1$ has at most two positive roots). We solved for $\alpha^*(q)$ independently by bisection:

q	second root $\alpha^*(q)$	interpretation
3	2.000000	exact
5	0.650919	
7	0.373501	
9	0.258108	

At $q = 3$ the second root is exactly $\alpha^* = 2$, matching the universal tail exponent for the classical Collatz reverse tree established by independent methods (Hill estimator on real-tree samples, and extreme value theory on block maxima) in earlier stages of this project. For $q \geq 5$, however, the second root falls *below* 1 rather than above it: a genuine structural transition, not a numerically small perturbation. Writing $\alpha_- < \alpha_+$ for the two roots, we have $\{\alpha_-, \alpha_+\} = \{1, 2\}$ at $q = 3$ but $\{\alpha_-, \alpha_+\} = \{\alpha^*(q), 1\}$ for $q \geq 5$: the trivial root becomes the *larger* of the two.

Theorem 3.6 (Structural transition: density, i.i.d. model)**.** *Let $N(x)$ be the population counting function of the i.i.d. branching-random-walk model whose annealed pressure is exactly $\rho_{\text{ann}}(\alpha) = q^{\alpha-1}/(2^\alpha - 1)$ (Theorem 3.3). Since $\alpha_-(q)$ is always unfrozen (Proposition 3.5),* $\log N(x)/\log x \to$ *$\alpha_-(q)$ almost surely, for every odd $q \geq 3$ [18]: exponent 1 (linear growth) at $q = 3$; exponent $\alpha_-(q) < 1$ for every $q \geq 5$.*

Proposition 3.7 (Finer asymptotic, conditional)**.** *Conditional on a renewal-type refinement, from the counting-function level (not merely the exponent) to the sharper asymptotic $N(x) \sim W \cdot x^{\alpha_-(q)}$, holding almost surely and uniformly,* including *at $q = 3$, for an almost surely positive, non-degenerate random scale factor W (not a deterministic constant at any q, since the underlying reproduction law has genuine variance). A directed literature search found the neighboring machinery but not this exact statement ready-made: the classical Crump–Mode–Jagers renewal theorem [19] is stated for one-sided ($[0, \infty)$) reproduction point processes, while our branching displacements* *include negative values (e.g. $2/3 < 1$ at $q = 3$, since a child can be numerically smaller than its parent); the branching-random-walk martingale theory that is natively two-sided [18] gives convergence generation-by-generation, not a renewal law summed over all generations; and the implicit renewal theory for weighted trees with general (two-sided) weights [14, 11, 10] resolves the* tail *of the fixed-point equation for W itself, not the cumulative counting law for $N(x)$ — a related but distinct question. We expect the present statement to follow from combining this machinery (spine decomposition and exponential tilting for a two-sided one-dimensional random walk, exactly as already used for the tail-of-W results just cited) but we have not carried out that adaptation, nor located it already written in this exact form; a very recent preprint [12] extends branching-process convergence laws to general type spaces without geometric rescaling and may already contain it, but we have not verified this. The non-lattice hypothesis that any such extension of [19] would need holds here, and it is precisely the irrationality of $\log q/\log 2$ that secures it. Although each single-step displacement $-\log q + a\log 2$ ($a \in \mathbb{Z}_{\geq 1}$) lies in one coset of the lattice $\log(2)\,\mathbb{Z}$, containment in a coset is not arithmeticity: the smallest closed subgroup of $\mathbb{R}$ generated by these displacements is $\log(q)\,\mathbb{Z} + \log(2)\,\mathbb{Z}$, which is dense in $\mathbb{R}$ for every odd $q \geq 3$ because $\log q/\log 2 \notin \mathbb{Q}$. Equivalently, the characteristic function of the displacement equals 1 only at the origin, so the renewal/CMJ asymptotic carries no log-periodic (period $\log 2$) modulation. Were $\log q/\log 2$ rational the branch positions would collapse onto a genuine lattice and a periodic correction would appear; the irrationality is therefore the operative fact, not an irrelevant one.*

Conjecture 3.8 (Structural transition: density, arithmetic tree)**.** *The same dichotomy holds for the genuine reverse tree of T_q itself, not merely for the i.i.d. model of Theorem 3.6 and Proposition 3.7: for $q = 3$ it has positive natural density ($N_u(x) \sim x$); for every odd $q \geq 5$ it has natural density zero, with $N_u(x) \sim W_u \cdot x^{\alpha_-(q)}$. At $q = 3$ this* implies, and strictly refines, *the Growth Exponent Conjecture of Kontorovich–Lagarias [6] and Applegate–Lagarias [20, 21] for the classical Collatz reverse tree, which asserts only the exponent $\pi_a(x) = x^{1+o(1)}$, not a specific asymptotic constant — open since 1995, with the best unconditional partial result remaining the one-sided bound $\pi_a(x) \geq x^{0.84}$ of Krasikov–Lagarias [16]. Our pressure identity reproduces the exponent-1 statement of this conjecture as the $\alpha_- = 1$ case of a single closed-form equation, together with its $q \geq 5$ analogue, rather than resolving it; the further claim of a specific limiting constant is additional content of Conjecture 3.8 beyond the classical conjecture.*

We state Theorem 3.6, Proposition 3.7, and Conjecture 3.8 separately, rather than as a single theorem about the arithmetic tree, precisely because the step from one to the other is exactly the kind of transfer that caused the error corrected in this section; Remark 3.9 below makes precise what does and does not rest on proof. Direct empirical support for Conjecture 3.8 is given in §3.4 below and by the statistically calibrated measurement of §6.

Remark 3.9 (What "transfers" rests on)*.* We are precise about the status of this transfer, since it is exactly the kind of step that caused the error corrected in this section. For the genuine i.i.d. branching-random-walk model that the annealed identity of Theorem 3.3 describes, quenched-equals-annealed growth throughout the unfrozen region is a rigorous, classical result [18]. For our arithmetic recursion, no analogous theorem is known to us: the gap is the same one already identified around Theorem 4.3 (Remark 4.4: near-independence of fresh digits across sibling subtrees, here in a quantitative large-deviations form rather than a qualitative one), and the closest genuine arithmetic result in the literature is a one-sided density bound of Applegate–Lagarias [20, 21], later sharpened by Krasikov–Lagarias [16] (already cited in §9), not a matching two-sided quenched statement. We therefore treat the transfer, at every root we invoke it for, as supported by (i) the classical stochastic-model theorem, (ii) direct numerical confirmation, and not by a proof for the arithmetic tree itself — naming the gap precisely rather than letting the stochastic-model theorem's rigor bleed by association into the arithmetic claim.

Conjecture 3.10 (Tail index of the scale factor)**.** *W_u has a regularly varying tail of index $\alpha_+(q)/\alpha_-(q)$, where $\alpha_+(q)$ is the larger root of (5) ($\alpha_+(3) = 2$; $\alpha_+(q) = 1$ for every $q \geq 5$).*

This is exactly the exponent predicted by the classical theory of fixed points of the (multitype) smoothing transform [22, 23, 24] for the genuine stochastic model:[1] the tail-index equation $\mathbb{E}[\sum_i A_i^s] = 1$ becomes, for our weights, $q^{\alpha-s-1} = 2^{\alpha-s} - 1$, solved exactly by $s = \alpha_+/\alpha_-$. This is further reason to take the conjecture seriously as a prediction, even though, per Remark 3.9, the theory only rigorously covers the stochastic model, not our arithmetic recursion.

We state this as a conjecture separate from both Theorem 3.6 and Conjecture 3.8 above, precisely because Proposition 3.5 shows $\alpha_+(q)$ is *always* frozen: the annealed pressure identity does not establish it for any q — not even at the level of the i.i.d. model of Theorem 3.6, whose Biggins-type transfer only covers the unfrozen root — and it must rest on evidence external to Theorem 3.3. At $q = 3$ that evidence is real and threefold: the Hill estimator on real-tree samples and extreme-value theory on block maxima (both from earlier stages of this project) give $\alpha \approx 2$ independently of the pressure argument, and the counting-exponent confirmation below adds a third, related line of support. For $q \geq 5$ the corresponding evidence is markedly weaker: an early Hill-estimator measurement of the tail of W_u for $q = 5$ (600 roots, agreement to two decimal places with the predicted $1.536\ldots$) was later shown, over the course of this project, to be statistically non-confirmatory at that sample size (the estimator's true standard error is ≈ 0.45, an order of magnitude larger than the agreement being claimed). We pursued this further along two independent routes, and report both honestly rather than stopping at whichever gave the more favorable number.

First, a substantially larger statistical test (5000 roots, four headroom levels up to 10^8; `collatz-endogeny/sec3-pressure-equation`) applied a battery of four estimators (a rank-size regression with the Gabaix–Ibragimov finite-sample correction; a bias-corrected Hill estimator following Huisman et al.; a generalized-Pareto threshold-stability scan; and a Clauset–Shalizi–Newman fit with a Vuong test against a truncated log-normal alternative). The outcome is mixed rather than confirmatory: the bias-corrected Hill estimator is remarkably stable across headroom levels and lands close to the predicted 1.536, but the threshold-stability scan shows no clean plateau near the predicted shape parameter, and the Vuong test favors the log-normal alternative over the power law, at conventional significance, in three of the four headroom levels tested. Reading the four estimators together by the depth of tail they each summarize shows the apparent local tail index rising smoothly with depth (from ≈ 1.3 at the widest window to ≈ 2.2 at the narrowest) rather than settling near any single value — consistent with a slowly-converging pre-asymptotic regime in which no estimator at this sample size can be treated as decisive.

Second, we tested the conjecture by an exact, non-statistical route: the same depth-k recursion $Z_k(\theta; u)$ used to verify the frozen-phase growth law above (with $\theta = \alpha_-(5)$) was evaluated over the *entire* population of residues $u \bmod 5^k$ (not a sample), for k up to 11 ($5^{11} \approx 4.9 \times 10^7$

[1] A related structural prediction of the multitype model, confirmed empirically for $q = 5$: conditioning on the root's residue class $i = u_0 \bmod q$ (with $a_0(i)$ the smallest admissible branch exponent for class i), the scale factor satisfies $W_i \overset{d}{=} 2^{-a_0(i)\theta} \cdot W^*$ for a common W^*, i.e. residue class only rescales W_u, it does not change its shape. Verified on 5000 sampled roots across four headroom levels, with quantile ratios matching this prediction to within 2–9%; exact only in the idealized i.i.d. model, as in Remark 3.9. The same relation survives unchanged when 2 is not a primitive root mod q (the extra sterile classes of §2): restricted to the surviving types $\langle 2 \rangle$, the multitype pressure matrix is rank 1, and its Perron eigenvalue collapses, by a d-independent cancellation (a_0 a bijection $\langle 2 \rangle \to \{1, \ldots, d\}$), to exactly $q^{s-1}/(2^s - 1)$ — the same pressure equation — so both the exponent and the scale-family direction survive intact; verified for $q = 7, 15$ to 1–4%. (Whether 2 is a primitive root mod q is, for prime q, governed by Artin's primitive root conjecture, open in general but proved conditionally on GRH by [17].)

residues; a sanity check — that the population mean of $Z_k(\theta; u)$ equals exactly 1 for every k, forced by Theorem 3.3 at $\alpha = \theta$ — passed to floating-point precision, confirming the implementation; `collatz-endogeny/sec3-pressure-equation`). If $W_u = \lim_k Z_k(\theta; u)$ has tail index ι, the population moment $M_k(p) = \text{mean}_u\, Z_k(\theta; u)^p$ should saturate in k for $p < \iota$ and diverge for $p > \iota$. The data show saturating behavior throughout $p \leq 1.6$ and diverging behavior for $p \geq 1.7$, which would place ι higher than the predicted 1.536 if taken at face value — but the ratio of successive increments has not yet stabilized for $p \leq 1.6$ at $k = 11$ (unlike $p \geq 1.7$, where it has), the classic signature of a system still relaxing toward its asymptotic exponent rather than one that has reached it. Given the previously observed slow convergence rate for $q = 5$ ($k^{-0.222}$ decay in the moment test — not an effect of the linear transfer operator's spectrum, which we later verified to be exactly $\{\Lambda, 0\}$ at every truncation level, i.e. a perfect spectral gap with no isolated subdominant eigenvalue; the transient's origin lies in a separate, nonlinear critical layer and remains open), the transient decays by only about 7% between $k = 8$ and $k = 11$; halving it would require $k \approx 250$, far beyond what exhaustive enumeration over 5^k residues can reach. We therefore cannot distinguish, from this data, between $\theta = 1.536$ lying just below a nearby true index and lying further below one — the test is inconclusive, not disconfirming, and enumeration alone will not resolve it.

Taken together, neither route confirms Conjecture 3.10 for $q \geq 5$ and neither refutes it; both are more informative than the original under-powered measurement, and both point to the same honest conclusion: this is a harder statistical question than the $q = 3$ case, already settled by the independent Hill/EVT measurements noted in §3. We have since pursued exactly the two follow-ups this pointed to. A pre-registered log-periodic fit (motivated by the branch weights being powers of 2) was tested against a period derived analytically, not fitted: the relevant renewal-theoretic dichotomy is non-arithmetic (the shift $\log_2 5$ is irrational), so no asymptotic log-periodicity is predicted, and a periodogram test found no power at the predicted frequencies. An analytic handle on the linear transfer operator's spectrum, once obtained, likewise turned out not to bear on this question, since the transient lives in a separate nonlinear layer (the operator has a provably perfect spectral gap). The other follow-up did move the needle: a pre-registered comparison on a $20\times$ larger sample (10^5 roots) produced, for the first time, a clean GPD threshold-stability plateau at the predicted value, a tightly concentrated Hill estimate, and a Vuong test that no longer favors the lognormal alternative – all confirmed against two synthetic calibration checks rather than taken at face value. This is now the strongest evidence to date in favor of Conjecture 3.10 for $q = 5$, though not a proof of it: Vuong non-rejection is not confirmation, and the martingale is provably still pre-asymptotic at the headroom reached. We list the conjecture as an explicit open problem in §13, updated to reflect this shift.

The mean drift $\log q - 2\log 2$ changes sign exactly at the threshold $q = 5$ separating the two cases of Conjecture 3.8: negative for $q = 3$ (orbits contract on average), positive for $q \geq 5$ (orbits expand on average). This is not a coincidence alongside the freezing computation of §3.2: at $\alpha = 1$ the entropy is exactly $s(1) = \log(4/q)$, so the same sign that decides contraction versus expansion of orbits also decides whether the trivial root $\alpha = 1$ is frozen.

3.4 Empirical confirmation on real reverse trees

Independent depth-first enumeration of the true (not extended) reverse trees for $q = 5$ and $q = 7$, following the exact admissibility rule of §2, provides direct empirical support for Conjecture 3.8: counting slopes per decade of 3 independently sampled roots gave ≈ 0.62–0.66 for $q = 5$ (predicted 0.650919) and ≈ 0.36–0.39 for $q = 7$ (predicted 0.373501). A more careful, statistically calibrated measurement for $q = 5$ is the subject of §6. As already noted above for $q = 3$ (where $w_2(1) = 1$ is a genuine fixed point), the smallest-branch map $u \mapsto w_{A_0(u \bmod q)}(u)$ has finite cycles for $q = 5$ (e.g.

$1 \mapsto 3 \mapsto 1$, since $w_4(1) = 3$ and $w_1(3) = 1$) and $q = 7$; naive enumeration would loop on nodes that reach one. Our implementation excludes all such cycle members explicitly (precomputed for each q) and deduplicates visited nodes, rather than relying only on sampling roots far from them.

Remark 3.11 (Bibliographic calibration). A directed novelty check against Kontorovich–Lagarias [6], Wirsching [4], and a later rigorous forward-direction result excluding $q \geq 5$ for the analogous $p = 2$ generalization [8] established that neither the numerical value $\alpha^*(5) = 0.650919$ nor the qualitative transition at $q \geq 5$ is new: Wirsching had already conjectured the transition heuristically in 1998, and Kontorovich–Lagarias had already computed the same quantity (denoted $\eta_{5,BP}$ in their Theorem 8.10) via large-deviations methods for a branching random walk, a route different from, but numerically identical to, our pressure-equation derivation. Given §3.2, this difference in route may matter more than a mere alternative derivation: a genuine large-deviations argument for a branching random walk is, in the appropriate sense, a quenched statement, whereas our pressure identity (Theorem 3.3) is annealed. That the two agree numerically at $\alpha_-(5)$ is consistent with $\alpha_-(5)$ being unfrozen (Proposition 3.5), where quenched and annealed are expected to coincide; it does not by itself certify that our annealed route would agree with a quenched one at a frozen root. What survives as new here is the closed-form equation (5) itself, unifying the family for arbitrary q in a single algebraic identity, together with its rigorous annealed derivation and independent numerical confirmation.

4 The endogeny barrier

4.1 Gauge freedom in the tree recursion

The recursion (2) is linear in G, and linearity has an immediate structural consequence that is easy to overlook: if W denotes the "arithmetic" solution — the one actually realized by, and measurable with respect to, the 3-adic digits of the tree — and Y is *any* bounded positive function invariant along the tree (in particular, any function of an auxiliary coordinate carried unchanged from parent to child), then $X := W \cdot Y$ solves *exactly the same recursion.*

Proposition 4.1 (Gauge freedom). *Let $\mathbb{Z}_3^* \times \mathcal{Y}$ be the product space on which the 3-adic digit dynamics acts on the first factor exactly as on the reverse tree, while passing an auxiliary coordinate $y \in \mathcal{Y}$ unchanged to every child. For bounded, non-constant $Y : \mathcal{Y} \to (0, \infty)$, the function $X(r, y) := W(r) \cdot Y(y)$ satisfies:*

- *(a) the same pressure equation* (5) *(identical weights, identical operator $M_q(\alpha)$, identical roots $\alpha = 1, \alpha^*$);*
- *(b) the same tail index as W;*
- *(c) the same linear regression slope $b \approx 1$ against the limiting measure μ, up to a fixed additional dispersion;*
- *(d) but* $\mathrm{Var}[\log X \mid$ *first m digits*$] \to \mathrm{Var}[\log Y] > 0$ *for* every *m, including $m = \infty$: the conditional variance never collapses.*

Item (d) is the crux: no statistic computed from pressure, tail index, or regression slope against μ distinguishes W from $W \cdot Y$. The technical name for this phenomenon is *endogeny* [7]: collapse of the conditional variance is equivalent to X being measurable with respect to the σ-algebra generated by the digits ("endogenous"); Proposition 4.1 exhibits a non-endogenous solution. We emphasize the logical status of this construction: it proves that the recursion *alone* does not force exactness; it does not assert that a non-trivial Y actually occurs for the arithmetic object built from a single integer v (there, Y would have to be manufactured from v itself, not injected as an independent auxiliary coordinate). These two statements must never be conflated.

4.2 A specific mechanism ruled out

One natural candidate mechanism for a non-trivial gauge factor is a persistent 2-adic "memory" of the root surviving deep into the tree. This is false:

Lemma 4.2 (No 2-adic memory)**.** *Let w be a child of v at exponent a in the reverse tree of the classical ($q = 3$) map, so $3w = 2^a v - 1$. Then*

$$w \equiv -3^{-1} \pmod{2^a},$$

a universal constant depending only on a, independent of v. Consequently, a descendant reached after a path with exponent sum A has its lowest A bits determined entirely by the path*, with zero information about the root.*

Proof. From $3w = 2^a v - 1$ we get $3w \equiv -1 \pmod{2^a}$, i.e. $w \equiv -3^{-1} \pmod{2^a}$ since 3 is a unit mod 2^a. Induction along the path, composing the relation at each generation, gives the claim. □

We verified Lemma 4.2 numerically for $a = 1, \ldots, 10$ against more than 500 values of v per exponent, confirming the predicted universal constant exactly in every case (e.g. $a = 5 \Rightarrow w \equiv 21 \pmod{32}$ always; $a = 10 \Rightarrow w \equiv 341 \pmod{1024}$ always). The reverse tree destroys all 2-adic information about the root at every generation; there is no persistent 2-adic coordinate available to correlate with the 3-adic residue, ruling out this specific mechanism for a gauge factor.

4.3 What forces collapse, and what does not

In the fully idealized probabilistic model — i.i.d. branching with independent subtree contents — exactness *is* forced, by two classical mechanisms:

(a) **Classification of smoothing-transform fixed points** [9, 10]: non-endogenous solutions of the associated smoothing-transform fixed-point equation generically have a strictly smaller tail index than the endogenous one (in the regime relevant here, tail index 1 with infinite mean), which is excluded by our own data: $\alpha^* = 2$ has been measured by three independent methods (pressure equation, Hill estimator, extreme-value theory on block maxima).

(b) **Choquet–Deny rigidity** for the Archimedean phase: bounded harmonic functions of a random walk on an abelian group are constant, which eliminates any Archimedean gauge coordinate under the full i.i.d. hypothesis.

Theorem 4.3 (The recursion does not force endogeny)**.** *The smoothing-transform model associated with the exact weight recursion of the reverse tree admits non-endogenous solutions (harmonic gauge factors, Proposition 4.1; the 2-adic obstruction is excluded by Lemma 4.2) with a strictly positive conditional-variance floor $\sigma_Y > 0$ at every resolution, statistically indistinguishable from the endogenous solution by pressure, tail index, or any marginal we have tested. Consequently the exact weight recursion together with the statistics of the 3-adic digits does not by itself force the residual gauge dispersion to vanish.*

Remark 4.4 (Barrier reading, and epistemic status). Informally this delimits a class of arguments: we are aware of no argument using only (i) the exact weight recursion and (ii) statistics of the 3-adic digits that does not factor through this smoothing-transform model, and hence none that can distinguish the endogenous from the non-endogenous solution. We do *not* formalize "uses only (i) and (ii)" as a precise class of techniques (in contrast to relativization or natural-proofs barriers in complexity theory), so this is a heuristic delineation of known approaches, not a formal impossibility theorem. The 2-adic exclusion (Lemma 4.2) and the gauge construction (Proposition 4.1) that Theorem 4.3 rests on are fully rigorous and independently verified; the classification claim above (that any such argument factors through the smoothing-transform model) can be stated unconditionally as "standard machinery under [hypotheses], detailed verification pending" rather than as independently re-derived against our exact setup. We never assert that the residual gauge dispersion σ_Y vanishes for the true arithmetic object; only that the recursion alone cannot force it to. The missing ingredient that closes the gap *in the idealized model* — and which remains to be transferred to the arithmetic setting — is near-independence, across sibling subtrees, of the fresh 3-adic digits of the leaf integers, together with equidistribution of the Archimedean phase along paths. The known rigorous analogue, in the forward direction, is the Fourier decay of Tao's Syracuse variables (Proposition 1.1); the reverse-tree/density version remains open and is *not* implied by it.

4.4 Precise regime of the tail exponent

A directed literature search (§11) raised the question whether $\alpha^* = 2$ coincides with the *critical case* of the classical multivariate smoothing-transform theory [13], characterized by $m'(\alpha) = 0$ at a root of $m(\alpha) = 1$ (a boundary/degenerate case in the sense of Biggins–Kyprianou). Writing $m(\alpha) := q^{\alpha-1}/(2^\alpha - 1)$ (the pressure function of §3 at $q = 3$), we compute

$$m'(1) = \ln 3 - 2\ln 2 \cdot \frac{2^1}{2^1 - 1} \quad \text{explicitly} \ \approx -0.288 < 0, \qquad m'(2) = \ln 3 - \tfrac{4}{3}\ln 2 \approx +0.174 > 0.$$

Since m is log-convex (as a sum, after clearing denominators, of exponentials in α), two distinct roots of $m(\alpha) = 1$ necessarily straddle the unique minimum of m, so *neither* can be a critical (double) root; criticality would require the two roots to merge, which does not happen at $q = 3$. The correct classification of $\alpha^* = 2$ is therefore as the *second root with* $m' > 0$ — a Goldie/Guivarc'h-type regularly-varying tail regime [14, 15], not the critical case. This distinction matters operationally: the critical-case uniqueness theorem of [13] *presupposes* exactly the independence between subtree copies that is the arithmetic gap of Theorem 4.3 — so even setting aside the regime mismatch, it could not have supplied a new tool for closing that gap. We note, tying this to §3.2, that "second root with $m' > 0$" is exactly the freezing condition of Proposition 3.5: the Goldie/Guivarc'h regime name correctly classifies the algebraic root $\alpha^* = 2$ of $m(\alpha) = 1$, but by itself says nothing about whether this root governs the *quenched* tail of our specific arithmetic recursion — which is exactly the question Proposition 3.5 and Conjecture 3.10 address directly.

5 Empirical calibration of the residual coupling

Proposition 4.1 shows the gap is real in principle; it does not quantify how large the residual coupling between sibling subtrees actually is for the true arithmetic tree. We designed a three-arm control experiment to calibrate this directly.

5.1 Design

Arm 1. **Synthetic i.i.d. control**: a Monte Carlo model with the exact branch weights $q \cdot 2^{-a}$ but genuinely independent subtree contents, truncated by total population size (not depth) to mirror the true enumeration procedure.

Arm 2. **Synthetic coupled control**: the same model, but with sibling subtree *contents* coupled at adjustable intensity $\rho \in [0, 1]$ via whole-family replication addressed by a hash of (phase, replicate id, tag), constructed so that all single-subtree marginals remain exactly unchanged for any ρ.

Arm 3. **Real arithmetic tree**: exact enumeration of the true reverse tree, re-measured at matched truncation depth.

The residual variance excess, (Arm 3)$-$(Arm 1), compared against the calibration curve floor(ρ) built from Arm 2, converts a qualitative statement ("the gap exists") into a quantitative bound on the effective coupling ρ_{eff} that would be needed in the synthetic model to reproduce whatever residual is observed in the real tree.

5.2 Result

Across the full truncation grid $m = 8, \dots, 29$, Arm 1 and Arm 3 showed nearly identical residual variances, with no sign of positive coupling. A final slope-and-bootstrap measurement at the deepest, best-calibrated truncation ($m = 20$, 4000 replicate roots, $\rho \in \{0, 0.05, 0.1, 0.2\}$) gives:

$$\rho_{\text{eff}} \lesssim 0.06 \quad (95\% \text{ CI}, \ m = 20). \tag{6}$$

The excess (real $-$ synthetic) converges monotonically to zero with increasing truncation depth, the opposite sign from what a genuine positive arithmetic coupling would produce, reinforcing rather than merely failing to contradict the conclusion. We state the scope of this bound precisely: no coupling of the specific functional form explored by Arm 2 (whole-family replication of subtree *contents* at intensity ρ) was detected in the real tree beyond $\rho_{\text{eff}} \lesssim 0.06$ of that family's maximum intensity. This is not, by itself, a bound on every conceivable form of coupling; in particular it does not bound the exact, coarse, Δ-dependent pair correlation of Proposition 10.2 below, which lives on a disjoint axis (root residue, not subtree content) already present identically, by construction, in Arm 1 itself — so it is not a form of coupling this calibration could have detected or missed. See Remark 10.3 for the verified structural argument. Equation (6) converts the purely theoretical gap of Theorem 4.3 into a quantitative, falsifiable empirical bound on its size, for this family of couplings.

6 An unconditional empirical result: resolving the Kontorovich–Lagarias vs. Volkov dispute for $5x + 1$

Independently of the theoretical program above, we report a fully unconditional empirical result of direct interest.

6.1 The dispute

For the reverse tree of T_5, Kontorovich–Lagarias [6] predict a counting exponent $\eta_{5,BP} \approx 0.650919$ (identical, as noted in Remark 3.11, to our pressure-equation root $\alpha_-(5)$), while a competing stochastic model due to Volkov, discussed in the same reference, predicts $\eta^*_{5,BP} \approx 0.678$. The authors of [6] themselves flag this as an open dispute, unresolved for lack of sufficiently powerful data; the two predictions differ by only $\Delta = 0.027$.

6.2 Method

We implemented the exact reverse tree of T_5 from scratch (the admissibility rule of §2 specialized to $q = 5$, $d = 4$: a child at exponent a exists from parent u iff $a \equiv A_0(u \bmod 5) \pmod 4$, with $A_0 = \{1 \mapsto 4, 2 \mapsto 3, 3 \mapsto 1, 4 \mapsto 2\}$; nodes with $u \equiv 0 \pmod 5$ are sterile). Unlike $q = 3$, there is no parity condition beyond integrality mod 5.

A first pilot (60 roots) already gave a slope of 0.6433 (95% CI $[0.6327, 0.6531]$), excluding 0.678 by roughly 6.4 standard errors. A larger production run ($n = 300$ roots) initially sampled roots too close to the measurement window's lower checkpoint, contaminating the truncation buffer; correcting this (roots restricted well below the window) gave a slope of 0.63801 (95% CI $[0.63159, 0.64410]$) — which, at this sample size, excluded *both* candidate values (0.678 by 12.6 standard errors, but also 0.650919 by 4.1 standard errors), a failure mode anticipated in advance: once the standard error falls below the residual truncation bias, a naive confidence interval on the biased estimator excludes everything, which is not evidence against the correct prediction.

We therefore rewrote the enumeration to track, in a single pass, the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13} (validated byte-for-byte against the earlier buffer-by-buffer method on four test roots). The pooled mean curve across roots decays geometrically:

$$0.60049\ (10^9) \to 0.62387\ (10^{10}) \to 0.63261\ (10^{11}) \to 0.63650\ (10^{12}) \to 0.63801\ (10^{13}),$$

with increments $0.0234, 0.0087, 0.0039, 0.0015$ shrinking by a factor of roughly 0.4–0.5 per decade of truncation — a signature of a genuine, extrapolable truncation bias, not noise.

6.3 Result

Aitken's Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \qquad 95\%\ \text{CI (bootstrap over roots)} = [0.633,\ 0.645]. \tag{7}$$

This excludes Volkov's prediction (0.678) with an ample margin (more than ten standard errors). The small residual gap to the Kontorovich–Lagarias value (0.650919, about 0.012 above our upper confidence bound) has a specific, diagnosable signature rather than being an unexplained discrepancy: the slope-per-decade panel *within* the fixed measurement window ($10^4 \to 10^5$ through $10^7 \to 10^8$) is still monotonically rising at the last decade tested ($0.6021 \to 0.6296 \to 0.6432 \to 0.6460$, no plateau), approaching 0.6509 monotonically without having reached it — the signature of *fixed-window pre-asymptotics* (the window is not yet "deep enough"), not of uncorrected truncation bias (already handled by the Aitken Δ^2 extrapolation above).

Empirical Result 6.1 (Empirical resolution of the Kontorovich–Lagarias vs. Volkov dispute)**.** *The* $5x + 1$ *reverse-tree counting exponent, measured by exact enumeration with Aitken-*Δ^2*-extrapolated truncation correction, is* 0.639 *(95% CI* $[0.633, 0.645]$*). This excludes the competing Volkov prediction (*0.678*) with high confidence and is consistent with the Kontorovich–Lagarias prediction (*0.650919*) up to a measured and explained finite-window pre-asymptotic effect.*

This result stands entirely on its own, independent of the theoretical program of §§4–10; we highlight it early in the paper's overall argument because it is the least framing-dependent contribution we report.

7 Direct computational test of the Weak Covering Conjecture

7.1 Wirsching's conjecture

Wirsching [4] identifies the following combinatorial covering property as sufficient for uniform positive predecessor density (his Corollary II.5.8 and Chapter V). Define

$$R_{j,k} := \left\{ \textstyle\sum_{i=0}^{j} 2^{\alpha_i} 3^i \ : \ j+k \geq \alpha_0 > \alpha_1 > \cdots > \alpha_j \geq 0 \right\} \subset \mathbb{Z}_3^*, \qquad |R_{j,k}| = \binom{j+k+1}{j+1}.$$

Every element of $R_{j,k}$ is automatically a unit mod 3 (its top-order term is not divisible by 3, every other term is), so "covering $\mathbb{Z}_3^*$ mod 3^ℓ" means hitting all $2 \cdot 3^{\ell-1}$ invertible residue classes.

Conjecture 7.1 (Weak Covering Conjecture; Wirsching 1998, Conj. 3.9)**.** *There is a sub-exponential function $K(\ell) > 0$ (i.e. $\lim_\ell K(\ell)e^{-\gamma\ell} = 0$ for every $\gamma > 0$) such that, for all j, ℓ: $|R_{j-1,j}| \geq K(\ell) \cdot 3^\ell \Rightarrow R_{j-1,j}$ covers $\mathbb{Z}_3^*$ mod 3^ℓ.*

For $j \geq \ell$, terms of index $i \geq \ell$ vanish mod 3^ℓ, giving the useful reduction $R_{j-1,j}$ mod 3^ℓ = $2^{j-\ell} \cdot R_{\ell-1,j}$ mod 3^ℓ, which cuts the enumeration cost from $\binom{2j}{j}$ to $\binom{j+\ell}{\ell}$.

7.2 Implementation and validation

We implemented a cyclic-rotation bitset dynamic program: processing candidate exponents in decreasing order, we maintain, for each count c of terms chosen so far, the set of reachable partial sums mod 3^ℓ as a single Python integer bitset; choosing the exponent v at position c is a cyclic rotation by $2^v \cdot 3^c$ mod 3^ℓ, and population count of the final bitset gives the image size directly. The implementation matched an independently computed reference table exactly ($j^*(\ell)$ for $\ell = 1, \ldots, 7$: $1, 4, 6, 7, 9, 10, 11$) and a pointwise check ($\ell = 2, j = 2$: image $= \{1, 2, 5, 7\}$ mod 9, missing $\{4, 8\}$).

7.3 Results

We computed $j^*(\ell)$, the smallest j for which $R_{j-1,j}$ covers, for $\ell = 1, \ldots, 20$ (cost grew by a factor ≈ 3.3 per step near the end; $\ell = 20$ alone took ≈ 27 minutes; further extension in pure Python was judged not worthwhile). Writing $e(\ell) := j^*(\ell) - \log_4 3 \cdot \ell$ for the excess over the entropy-matched threshold $\log_4 3 \approx 0.7925$:

ℓ	j^*	j^*/ℓ	$e(\ell)$
1	1	1.0000	0.208
5	9	1.8000	5.038
10	15	1.5000	7.075
15	20	1.3333	8.113
16	20	1.2500	7.320
17	21	1.2353	7.528
18	22	1.2222	7.735
19	23	1.2105	7.943
20	24	1.2000	8.150

(Full table for all 20 values is in the accompanying computational record.) $j^*(\ell)$ exists and is finite throughout the tested range. An earlier reading of the first 17 points suggested a monotonically decelerating local slope approaching the threshold $\log_4 3$, which the three new points at $\ell = 18, 19, 20$ (all increments exactly $+1$, no new plateaus) revealed to be an artifact of an isolated plateau at $\ell = 16$ crossing the sliding measurement window: without it, the local slope oscillates in 0.74–0.86, straddling $\log_4 3$ from both sides rather than converging cleanly.

Model comparison on the tail $\ell = 10$–20 (constant vs. logarithmic vs. square-root vs. slow-linear growth of $e(\ell)$, by AIC) now disfavors pure stabilization ($\Delta\text{AIC} \approx 5$), consistent with a plateau-frequency test (only 1 plateau observed in 19 increments versus ≈ 3.9 expected under stabilization at rate $\log_4 3$; binomial $p \approx 0.072$, marginal but in the same direction). The four slow-growth models remain statistically indistinguishable from one another in this range ($\Delta\text{AIC} < 0.5$); they would only separate by ≥ 1 unit of j^* near $\ell \approx 40$, out of reach of this algorithmic approach.

Empirical Result 7.2 (Direct computational test of the Weak Covering Conjecture). *$j^*(\ell)$ exists and is finite for all $\ell \leq 20$ (the first direct computational test, to our knowledge, of the exact object of Conjecture 7.1). The excess $e(\ell)$ qualitatively favors unbounded slow growth over stabilization (two independent statistics, ΔAIC and plateau frequency, in the same direction but not individually decisive), consistent with the weak form (Conjecture 7.1) and disfavoring the corresponding strong form with $K(\ell)$ bounded.*

8 Three precision regimes for Tao's bivariate extension

8.1 Why the naive bivariate argument fails

A natural strategy for closing the gap of Theorem 4.3 is to apply Proposition 1.1 twice, once per sibling leaf, and combine via some form of conditional independence given the common ancestor. A close reading of the mechanism of Tao's Section 7 proof (characters χ_ξ, i.i.d. Pascal-distributed valuation pairs, a phase evolving multiplicatively by $\log 9/\log 2$, a "black set" of well-separated triangles, a two-dimensional renewal process with monotonicity) shows that this mechanism operates entirely on $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ as constructed from *independent* geometric digits by definition (1); the link to actual integers is made elsewhere in [1] (via a total-variation approximation valid only for depth $n \lesssim \log N$).

For our problem — a single fixed integer v, two sibling leaves $w_1(v), w_2(v)$ — the two leaves are *not* two i.i.d. samples; they are two different deterministic functions of the *same* variable v.

Proposition 8.1 (Perfect dependence at fixed depth). *For a fixed pair of sibling paths of depth D, the existence congruence pins v to a single class mod 3^D: writing $v = v_0 + 3^D t$, the two leaves become affine functions of the same free variable t, with unit multipliers: $w_i = w_i(v_0) + 2^{A_i} t$. Consequently the pair $(w_1, w_2) \bmod 3^\ell$ lies on a line of size 3^ℓ inside $(\mathbb{Z}/3^\ell\mathbb{Z})^2$, for every $\ell \geq 1$ — perfect deterministic dependence at every precision, with an entire line of resonant frequencies where the correlation has modulus exactly 1.*

Consequently, "apply Proposition 1.1 twice, assuming conditional independence given the ancestor" is not merely circular: it is false as stated, at any precision.

8.2 The correct, non-circular reformulation

The quantity that actually matters — decorrelation of *subtree aggregates*, not leaf-to-leaf — is a second-moment statement, expanding into a sum over *pairs of paths*:

$$\mathbb{E}_v[Z_1 Z_2] \;\propto\; \#\{(a, a') : a_1 \in B_1,\; a'_1 \in B_2,\; \mathrm{Syrac}(a) \equiv \mathrm{Syrac}(a') \bmod 3^D\},$$

which by Fourier inversion gives $\mathrm{Cov} \propto \sum_{\xi \neq 0} S_1(\xi) S_2(\xi)^*$, with S_i character sums conditioned on the first step. Here there is no circularity: the two path indices are independent by the tautology of expanding a square, not by an arithmetic hypothesis.

8.3 The three regimes

1. **Fixed pair, any ℓ**: perfect dependence (Proposition 8.1); no decay is possible, and decorrelation can only come from averaging over paths, never leaf-to-leaf.

2. $\ell = O(\log D)$: *provable*, with Tao's Section 7 machinery essentially unchanged. The budget $|\rho(\xi)| \leq C_A D^{-A}$ closes the sum over $3^\ell - 1$ frequencies provided $3^\ell \ll D^{2A}$, i.e. polynomial moduli in D — an honest, reachable lemma, though its most natural formulation turns out to be false as originally sketched (§10).

3. $\ell \asymp c \cdot D$ (where the endogeny barrier, the fresh digits, and Wirsching's Weak Covering Conjecture all live): this requires *exponential* (power-saving), ξ-uniform Fourier decay; Proposition 1.1 gives only super-polynomial decay, and Tao's Section 7 method structurally cannot give more, because the losses in the black-set argument are tied to Diophantine properties of $\log 3/\log 2$ via the slope $\log 9/\log 2$.

Theorem 8.2 (Regime 3 is at least as hard as recognized open problems)**.** *Regime 3 above is a special case of, and in the generality required here is at least as hard as, each of: (i) correlations of digital functions in bases 2 and 3 [30], (ii) effective rigidity for $\times 2, \times 3$-invariant structures in the sense of Furstenberg (the pair (w_1, w_2) is v observed through two elements of the affine semigroup generated by $x \mapsto (2^a x - 1)/3$), open in the generality required here, and (iii) exponential Fourier decay for 3-adic self-similar measures of Breuillard–Varjú type, known only in special algebraic cases. None of these furnishes a usable tool today.*

9 Triangulation: three independent formulations of the same obstruction

9.1 $\beta = 1$ (Tao 2020) is the Weak Covering Conjecture (Wirsching 1998)

Tao [3] defines $c_n := \inf_{b \in \mathbb{Z}/3^n\mathbb{Z},\, 3 \nmid b} \Pr(\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) = b)$ and proposes the conjecture $\beta = 1$: $c_n = 3^{-n+o(n)}$ ("no residue class has probability much smaller than average"), proving conditionally that $\beta = 1$ implies Collatz density $\gg x^{1-o(1)}$ (almost total), and noting that the best known unconditional bound is Krasikov–Lagarias' $\gamma = 0.84$ [16].

Proposition 9.1 (Structural identity)**.** *After a unit twist, Tao's Syracuse random variable and Wirsching's sets $R_{j,k}$ are the same algebraic object (sums of powers of 2 and 3 with strictly decreasing exponents). Consequently, Conjecture 7.1 (sub-exponential $K(\ell)$) implies $\beta = 1$ (an exact entropy count: $j = \log_4 3 \cdot \ell + o(\ell)$ gives $c_\ell \gtrsim 3^{-\ell(1+o(1))}$); conversely $\beta = 1$ alone gives only an exponentially weakened form of Conjecture 7.1. The two are sibling formulations on the same exact entropy scale — the threshold $\log_4 3 \approx 0.7925$ of §7 is the same identity 4^j vs. 3^ℓ on both sides, not a coincidence.*

Corollary 9.2. *The excess $e(\ell) = j^*(\ell) - \log_4 3 \cdot \ell$ measured in §7 is literally the $o(n)$ term of $\beta = 1$ ($c_\ell \gtrsim 3^{-\ell} \cdot 4^{-e(\ell)}$): our qualitative reading of $e(\ell)$ (sub-linear growth favored) is direct computational evidence in favor of $\beta = 1$, not merely of Conjecture 7.1.*

Tao's 2020 approach also shows that the endogeny/decorrelation route of §4 is not the only path: it replaces "near-independence between subtrees" with "deterministic separation of preimages (easy) plus L^∞ marginal control ($\beta = 1$, where all the difficulty lives)". This confirms that a route avoiding decorrelation altogether exists, and that it hits the *same* wall in marginal form.

9.2 Wirsching's 2003 functional equation and the base-3 Fabius function

A different, more refined route toward the same goal is Wirsching [5], whose central combinatorial object, the "Elka" functions $e_\ell(k,a) := |E_{\ell,k}(a)|$, count paths of length $k+\ell$ in the Collatz graph ending at a — the same counting object underlying our entire $G(v)$/Syrac program, now viewed on the state space $\mathbb{X} = \mathbb{I}_3 \times \mathbb{Z}_3^\times$: the $\mathbb{Z}_3^\times$ factor carries the 3-adic variable (habitat of the Syracuse measure), while the $\mathbb{I}_3 \cong \{0,1,2\}^{\mathbb{N}}$ factor carries the normalized step-count $k/3^\ell$, converging to a density φ.

Proposition 9.3 (Base-3 Fabius function)**.** *φ is the density of $X = \sum_{j\geq 1} 2U_j \cdot 3^{-j}$, U_j i.i.d. uniform on $[0,1]$ — the base-3 analogue of the Fabius function (which satisfies $f'(x) = 2f(2x)$; here $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $[0,2/3]$), a member of the family of "atomic functions" of Rvachev. It is the unique $L^1([0,1])$ fixed point of the averaging operator $W_3 f(x) := \frac{3}{2}\int_{3x-2}^{3x} f(t)\,dt$, C^∞, piecewise polynomial away from the standard Cantor set, with certified geometric convergence $\|f_n - \varphi\|_1 \leq 2^{-n+1}\|f_1 - f_0\|_1$.*

Wirsching's chain of implications reduces uniform positive predecessor density to a sequence of three conjectures, the most concrete of which (Conjecture 3, an inequality on $\liminf \varphi(z_\ell)/\varphi_0(z_\ell)$ along sequences in the central-limit window $|\ell - k_\ell| \leq \delta\sqrt{\ell}$) we tested numerically.

9.3 Certified numerical test of Wirsching's Conjecture 3

Using the exact self-similarity $X \overset{d}{=} (2U + X)/3$, the moments $M_i = \mathbb{E}[X^i]$ satisfy the exact rational recursion

$$M_i = \frac{1}{3^i - 1}\sum_{k=1}^{i}\binom{i}{k}\frac{2^k}{k+1}M_{i-k},$$

and, combined with an antiderivative reduction expressing φ at extreme-tail arguments in terms of exact binomial moment sums (rather than iterating W_3, whose L^1 error certificate is too weak to control pointwise values at the relevant scale), we obtain a zero-truncation-error evaluation of φ at depth ℓ up to $\ell = 500$ (working throughout in log-space with 60-digit precision; error certified $\leq 10^{-8}$).

Empirical Result 9.4 (Numerical test of Wirsching's Conjecture 3)**.** *The decisive ratio $L_\ell := 3^{1-\ell}\varphi(3x_\ell^+)/\varphi(x_\ell^+)$, measured for $\ell = 250,\ldots,500$ in the CLT window ($u \in [-2,2]$), converges to the predicted value $2/3$ with deficit $\ell\cdot(2/3 - L_\ell) \to 0.580 \pm 0.001$ — a coefficient independently reproduced by Berg–Krüppel's own asymptotic φ_0. The quantity $\ln(\varphi/\varphi_0)$ is increasing, concave, and uniform across $u \in [-2,2]$ (dispersion $< 10^{-4}$), consistent with a finite limit $L = -0.619 \pm 0.001\,(\text{stat})$, with a tail-shape systematic uncertainty of ± 0.015, giving $c = e^L \approx 0.538$ (honest range 0.53–0.55), best fit by a tail term $0.79/\sqrt{\ell}$; no log-periodic modulation is detected. This supports Conjecture 3 (Wirsching's most concrete conjecture in the chain) with certified numerical error, on the tested range; Conjectures 1 and 2 above it remain open.*

Together, Empirical Results 7.2 and 9.4 and Proposition 9.1 give three independent formal "guises" — integer counting (Wirsching 1998), Fourier characters (Tao 2020), a real functional equation (Wirsching 2003) — of the same critical window, none resolved.

10 Two candidate lemmas, executed to completion

Sections 8–9 identify two natural next steps: a "regime-2 lemma" (decorrelation of subtree aggregates at logarithmic precision) and a "Z-number reduction lemma" (formalizing the analogy with

Littlewood–Offord/Bohr-set methods from [2]). We executed both to completion; neither survives as originally sketched, but both yield genuine, citable results.

10.1 The regime-2 lemma: refutation, structure theorem, and a computational refutation of its natural sufficient condition

The originally sketched budget, $|\rho(\xi)| \leq C_A D^{-A}$ *uniformly in* ξ, contradicts Proposition 1.2 (an exact identity, not a lossy estimate): for any frequency ξ of primitive level ℓ' (i.e. $\xi = 3^{\ell-\ell'}\xi_0$, $3 \nmid \xi_0$) and any depth $D \geq \ell'$, $\mathbb{E}[e(\xi F_D/3^\ell)] = \varphi_{\ell'}(\xi_0)$ is *independent of* D: depth beyond ℓ' is a dummy parameter, because 3-adic ultrametricity truncates the relevant sum to the coarsest ℓ coordinates (Proposition 1.2). A closed example at $\ell' = 1$: $\mathrm{Syrac}_1 = 2^{-a} \bmod 3 \in \{1, 2\}$ with probabilities $1/3, 2/3$; $|\varphi(1)| = 1/\sqrt{3} \approx 0.577$ for *every* D, with no decay whatsoever. The correct per-frequency budget is $C_A \ell'^{-A}$ (Proposition 1.1, which requires $3 \nmid \xi$), not $C_A D^{-A}$; even with this correction, the total ℓ^∞ budget summed over frequencies, $\sum_{\ell' \leq \ell} 3^{\ell'}(C_A \ell'^{-A})^2 \asymp 3^\ell \ell^{-2A}$, never tends to 0: closing by frequency counting would require square-root cancellation, $\ell^{-\ell'/2-\varepsilon}$ decay — Regime 3 resurfacing inside what should have been the easy regime.

What survives is a genuine structure theorem. With $w_2 = 2^\Delta w_1 + c_\Delta$, $c_\Delta = (2^\Delta - 1)/3$ (a unit multiplier mod 3^ℓ) relating admissible sibling leaves at exponent gap Δ:

Lemma 10.1 (Covariance spectral identity)**.** *For subtree-aggregate observables $Z_i := N_{\eta_i}^{(m)}(w_i(v))$ (weighted populations truncated to precision m), with v uniform on admissible classes mod 3^{m+1},*

$$\mathrm{Cov}(Z_1, Z_2) = 3^{-2m} \sum_{\xi \neq 0 \bmod 3^m} e(-\xi c_\Delta/3^m)\, S_1(-2^\Delta \xi)\, S_2(\xi),$$

by character orthogonality and the affine sibling relation. The two summation indices are independent by the tautology of expanding the square (not an arithmetic hypothesis), but the dependence through the common v is not thereby cancelled: it survives entirely as the diagonal frequency pairing $\xi_1 = -2^\Delta \xi_2$.

Proposition 10.2 (Exact coarse endogeny)**.** *For complete populations ($\eta_i \equiv 1$) and precision $\ell = 1$: $\mathrm{Corr}(Z_1, Z_2) = +1$ if $\Delta \equiv 0 \pmod 6$, and $= -1/2$ otherwise — independent of depth D and of any truncation.*

Proposition 10.2 converts the endogeny barrier of Theorem 4.3 from a qualitative argument into a closed form: aggregating leaves does *not* decorrelate them; aggregation projects onto the coarse σ-algebra generated by $w_i \bmod 3^\ell$, where the affine sibling relation of the fixed-pair regime (§8, Regime 1) reappears intact rather than vanishing. This is a positive result with the opposite sign from what was hoped for: what is provable is the persistence of coarse correlation, not its disappearance.

Decomposing the sum of Lemma 10.1 by primitive level and applying Cauchy–Schwarz plus Parseval gives $\mathrm{Cov} = C_{\mathrm{exact}}(\ell_0; \Delta) + \mathrm{Err}$, with C_{exact} exact, finite, and D-independent (Proposition 10.2), and

$$\frac{|\mathrm{Err}|}{\mathbb{E}[Z_1]\mathbb{E}[Z_2]} \;\leq\; K_m - K_{\ell_0}, \qquad K_\ell := 3^\ell \sum_y \mu_\ell(y)^2,$$

the normalized *collision mass* of the Syracuse measure μ. A conditional lemma would follow if $K_\infty := \lim_\ell K_\ell < \infty$ (equivalently, the density $f = d\mu/d(\mathrm{Haar})$ lies in $L^2(\mathbb{Z}_3)$): the fine covariance, net of the exact coarse component, would then be $\leq K_\infty - K_{\ell_0} \to 0$, uniformly in m, D, Δ.

Remark 10.3 (Consistency with the calibration of §5). Proposition 10.2 and the null result $\rho_{\text{eff}} \lesssim 0.06$ of §5 measure different objects and do not conflict, though the point is worth making explicit. The sharpest reason is structural, not merely quantitative: in the synthetic model underlying §5's Arm 1 and Arm 2, a child's type (its residue mod 3 — exactly the $\ell = 1$ object Z_i of Proposition 10.2) is computed as a deterministic function of the root's phase and the branch index alone, reproducing the true cyclic admissibility structure (validated against real integers), and is entirely independent of the coupling intensity ρ that Arm 2 varies: we checked this directly in the simulator's code (2000 seeds, $\rho \in \{0, 0.3, 0.7, 1\}$, zero divergence in the resulting type sequence), and confirmed that this deterministic type structure reproduces the exact correlation of Proposition 10.2 ($+1$ for $\Delta \equiv 0 \pmod 6$, $-1/2$ otherwise) to four decimal places. In other words, the coarse correlation is already present, identically, in Arm 1 ($\rho = 0$, the "no coupling" null) as it is in the real tree; ρ instead governs a disjoint axis — whether sibling subtrees' *content* (the fresh randomness setting each subtree's *size*) is shared, not whether their root residues are correlated. The bound $\rho_{\text{eff}} \lesssim 0.06$ therefore could not have detected, or failed to detect, Proposition 10.2's correlation in the first place: that correlation was never a form of coupling the calibration's null hypothesis removes. A secondary, quantitative reason reinforces this: even granting the coarse correlation full weight, its contribution to the *aggregate* variance the calibration measures is diluted. Proposition 10.2's covariance $C_{\text{exact}}(\ell_0; \Delta)$ is finite and D-independent, while the aggregate means $\mathbb{E}[Z_i]$ grow exponentially with truncation depth, so its normalized contribution $C_{\text{exact}}/(\mathbb{E}[Z_1]\mathbb{E}[Z_2]) \to 0$ as depth increases — consistent with the excess of §5 decaying monotonically to zero with truncation depth. Separately, the refutation of the L^2 hypothesis below (Theorem 10.4) is the failure of an *upper bound* on the fine covariance ($K_\infty = \infty$ makes the bound above vacuous), not a proof that the fine covariance is in fact large; §5 measures directly the fine coupling this bound leaves uncontrolled, and finds it small. Finally, the geometric branch weights $q \cdot 2^{-a}$ concentrate on small gaps Δ: weighting pairs of branches by their natural second-moment mass $2^{-a_i} \cdot 2^{-a_j}$ (the same weighting entering Cov above), $\Delta = 2k$ has probability $3 \cdot 4^{-k}$, giving $\Pr[\Delta \equiv 0 \pmod 6] = 1/21$ against $\Pr[\Delta \not\equiv 0 \pmod 6] = 20/21$ exactly, so the mean coarse correlation under this natural weighting is exactly $\mathbb{E}[\text{Corr}] = \frac{1}{21}(+1) + \frac{20}{21}(-\frac{1}{2}) = -\frac{3}{7}$ (verified in `collatz-endogeny/sec10-l2-refutation-and-jensen/`), consistent with the observed negative sign of the residual excess in §5, though (per the structural point above) no such quantitative match is actually required to reconcile the two results. The two sections are therefore complementary rather than in tension: Theorem 4.3 and Proposition 10.2 establish that the recursion *cannot force* sibling decorrelation (the per-pair coarse structure survives at every depth, in every model that respects the true admissibility rule), while §5 establishes that the actual aggregate residual, on the orthogonal axis of subtree *content*, is nonetheless empirically small.

Theorem 10.4 (Computational refutation of the L^2 hypothesis)**.** *K_ℓ was computed exactly via the recursion*

$$\mu_n(y) = \tfrac{1}{2}\,\nu(2y \bmod 3^n) + \tfrac{1}{2}\,\mu_n(2y \bmod 3^n), \tag{8}$$

where ν is the law of $1 + 3F_{n-1}$ (derived below), solved as a truncated geometric series along the cyclic orbit of multiplication by 2 mod 3^n (a single orbit, since 2 is a primitive root mod 3^n for every n). For $\ell = 1, \ldots, 17$, K_ℓ grows cleanly and consistently linearly*, with increments converging monotonically to ≈ 0.47, with no sign of saturation ($K_1 = 5/3$ exactly, matching a hand computation from* $\Pr(\text{Syrac}_1 = 1) = 1/3$, $\Pr(\text{Syrac}_1 = 2) = 2/3$*). Hence $K_\infty = \infty$: the L^2 hypothesis fails.*

Derivation of the recursion (8). By Proposition 1.2, $F_n(a) = 2^{-a_1}\big(1 + 3F_{n-1}(a_2, \ldots, a_n)\big) \bmod 3^n$. Write $X := 1 + 3F_{n-1}(a_2, \ldots, a_n)$, with law ν, independent of a_1. Conditioning on $a_1 = 1$ (probability $1/2$) gives $F_n = 2^{-1}X$; conditioning on $a_1 \geq 2$ (probability $1/2$), the memorylessness of the geometric distribution gives $a_1 - 1 \sim \mathrm{Geom}(2)$ again, so $F_n = 2^{-1}\big(2^{-(a_1-1)}X\big)$, and the bracketed quantity has exactly the law of F_n itself (an independent copy). This gives the self-referential fixed point $\mu_n(y) = \frac{1}{2}\nu(2y) + \frac{1}{2}\mu_n(2y)$, all arithmetic mod 3^n. $\square$

We validated the recursion — not merely its $\ell = 1$ base case, which would not exercise the recursive machinery at all — against direct Monte Carlo sampling of Syrac from its primary definition (1) at $\ell = 3, 4$ (3×10^6 samples each), bin-by-bin: maximum discrepancy 0.000311 ($\ell = 3$) and 0.000213 ($\ell = 4$), both within the expected Monte Carlo noise ($\approx 1/\sqrt{N} \approx 0.000577$).

Remark 10.5 (Connection to the tail exponent). H-104 (an earlier stage of this project) measured a tail exponent $\alpha \approx 2$ for subtree-population ratios — exactly the critical exponent of the L^2 condition ($\alpha > 2 \Rightarrow K_\infty < \infty$; $\alpha < 2 \Rightarrow K_\ell$ diverges polynomially; $\alpha = 2$ is the borderline case). Theorem 10.4's clean linear growth is consistent with α slightly below 2, or exactly at the threshold with a logarithmic correction.

Corollary 10.6 (A ladder of "flatness" for the Syracuse measure)**.** L^1 *(proved, Tao's Prop. 1.14:* $\mathrm{Osc}_{m,n} \ll_A m^{-A}$*, giving total-variation approximability by absolutely continuous measures)* $\succ$ L^2 *(refuted here, Theorem 10.4)* $\succ$ $L^\infty/\beta = 1$ *(§9, open) — neither comparable to nor implied by* L^2*, but living on the same "beyond total-variation" floor of the ladder.*

The regime-2 lemma is thus not an easier stepping stone toward Regime 3: it is a sibling of it, sharing the same missing ingredient, now with a direct computational refutation of the natural sufficient condition rather than mere silence.

10.2 The Z-number reduction: a conditional dichotomy and an exact negative result

The second candidate lemma formalizes "failure of the Weak Covering Conjecture at scale ℓ with margin c implies a Bohr-type concentration configuration", via the Littlewood–Offord/Riesz-product technique of Tao's 2011 blog post [2] (not a paper; we flag this bibliographic correction explicitly, since it is a recurring source of miscitation). That post proves conditional separation results for $2^a - 3^k$ via Halász's method [32] and inverse Littlewood–Offord theory [33, 34], but acknowledges that Baker's theorem already gives an unconditionally stronger separation, so treats its own propositions as methodological exercises. Mahler's classical "Z-number" problem [35] concerns $\xi(3/2)^n$, a genuinely different (though structurally adjacent) object from ours (which concerns $\times 2 \bmod 3^s$ at finite depth, not $\times 3/2 \bmod 1$ on an infinite orbit); the honest anchor for our object is the Erdős/Lagarias family [37, 38].

Definition 10.7 (Failure with margin c)**.** For $c > 0$, the Weak Covering Conjecture *fails at scale* ℓ *with margin* c if, for $j = \lceil (1+c)\log_4 3 \cdot \ell \rceil$, some $b \in (\mathbb{Z}/3^\ell\mathbb{Z})^\times$ lies outside the image of $R_{j-1,j}$. Set $\gamma_c := (j+\ell)/\ell = 1 + (1+c)\log_4 3$.

Definition 10.8 (Diagonal configuration)**.** A configuration of depth s at slope γ, tolerance δ, and exception rate η is $\xi \in \mathbb{Z}$, $3 \nmid \xi$, such that in $\geq (1-\eta)s$ of the scales $t \in \{1, \ldots, s\}$ there is $a \in [\gamma t - t^{2/3}, \gamma t + t^{2/3}]$ with $\|\xi \cdot 2^a/3^t\| < \delta$.

Lemma 10.9 (Pre-wrap triviality)**.** *For* $\xi = 1$, $\|2^a/3^s\| < \delta$ *holds trivially for* $a \leq \log_2 3 \cdot s + \log_2 \delta$ *(the representative has not yet wrapped around). Since* $\gamma_c \geq 1.7925 > \log_2 3 \approx 1.585$ *for every* $c \geq 0$*, the forced configuration lives entirely in the non-trivial post-wrap regime, by structural slack coming from the* 4^j *vs.* 3^ℓ *entropy accounting.*

Definition 10.10 (Spectral concentration $SC(\varepsilon)$)**.** $\mu_{\ell,j}$ (the law of $Y = \sum R \bmod 3^\ell$) satisfies $SC(\varepsilon)$ if there is $\xi \not\equiv 0 \bmod 3^\ell$ with $|\hat{\mu}_{\ell,j}(\xi)| \geq 3^{-\varepsilon\ell}$.

Lemma 10.11 (Conditional reduction)**.** *For $c > 0$ and $\varepsilon < \varepsilon_0(c)$ there exist $\delta(\varepsilon, c) < 1/2$, $\eta(\varepsilon, c) < 1$ such that: failure at scale (ℓ, c) together with $SC(\varepsilon)$ implies existence of a (γ_c, δ, η)-diagonal configuration of depth $s \geq (1 - O(\varepsilon))\ell$. Contrapositively: absence of diagonal configurations of slope γ_c at large depth implies any failure of the Weak Covering Conjecture, if it exists, is spectrally* diffuse *($SC(\varepsilon)$ fails for every ε).*

The proof (a six-step argument: Fourier-inversion lower bound on $\sum |\hat{\mu}(\xi)|$, transfer to a tilted i.i.d. model at parameter $p_c = 1/\gamma_c$, extraction of alternable parallelepipeds via the Riesz-product device of [2], Halász's theorem, a routine scale descent, and an upgrade from a diagonal configuration to a continuous segment) is complete except for the last step, which requires stability of the relevant frequency ξ under $j' \mapsto j' + 1$ across scales — plausible but not established, and declared as an open gap rather than glossed over.

10.3 Proposition C: an exact wall of constants

Proposition 10.12 (Halász deficit)**.** *Unconditionally, Halász's theorem gives only $\tau \geq 3^{-\ell}$, not $SC(\varepsilon)$, yielding a budget $\sum \|\cdot\|^2 \leq 0.549\ell$ against a required ceiling of 0.25ℓ ($\delta = 1/2$, $d = \ell$ generators): a deficit of a factor ≥ 2.2, rising to ≈ 7 at a realistic generator density $\rho \approx 0.3$. In Tao's original 2011 application this closes because $\log q \ll d$ (unbounded ratio); in the Weak Covering Conjecture regime, $\log_2 q/d \geq 1.6$ — the technique does not merely degrade, it inverts sign.*

Theorem 10.13 (Jensen identity and the exact Fourier-budget deficit)**.** *Let $Z = \sum_{g \geq 1} w_g\, e(U_g)$, U_g i.i.d. uniform, $w_g = p(1-p)^{g-1}$ ($\sum_g w_g = 1$), with $p \geq \frac{1}{2}$. Then, exactly,*

$$\Lambda := \mathbb{E}\big[\log(1/|Z|)\big] = \log(1/p). \tag{9}$$

Proof. By the memoryless self-similarity of the geometric weights ($w_g/(1 - w_1) = p(1-p)^{g-2}$ for $g \geq 2$, the same geometric sequence re-indexed), $Z \overset{d}{=} p\, e(U_1) + (1 - p)\, Z'$, with Z' an independent copy of Z; since $|Z'| \leq 1$ always (a convex combination of unimodular terms), and $p/(1 - p) \geq 1$ for $p \geq 1/2$, we have $|Z'| \leq p/(1-p)$ whenever $p \geq 1/2$. Writing $c := (1-p)Z'/p$, so $|c| \leq 1$, factor $p\, e(U_1) + (1-p)Z' = p\, e(U_1)\big(1 + c\, e(-U_1)\big)$ and take $\log|\cdot|$: the function $z \mapsto \log|1 + cz|$ is (the real part of) a holomorphic function of z with no zero inside the disc $|z| < 1/|c| \ni$ the unit circle (as $|c| \leq 1$), so by the mean-value property (Jensen's formula with no zeros enclosed) its average over $z = e(U_1)$, U_1 uniform, equals its value at $z = 0$, namely 0. Hence $\mathbb{E}_{U_1} \log|p\, e(U_1) + (1-p)Z'| = \log p$ for every realization of Z', and averaging over Z' gives $\mathbb{E}[\log|Z|] = \log p$, i.e. (9). □

Applying Theorem 10.13 with $p = p_c = 1/\gamma_c \approx 0.558$ (the tilted model matching the Weak Covering Conjecture's threshold slope; note $p_c \geq \frac{1}{2}$, as the theorem requires) gives $\Lambda = \log \gamma_c \approx 0.5836 + O(c)$, confirmed by Monte Carlo (8×10^6 samples: 0.58344 ± 0.0005), against the value $\log 3 \approx 1.0986$ that would be needed to exclude a diffuse hole by an annealed ℓ^1 Fourier-budget argument: a deficit factor of ≈ 1.88. The annealed criterion would only invert at $\gamma = 3$ ($j \geq 2\ell$), far above the regime of the conjecture ($j^*(\ell)/\ell \approx 1.2$, Empirical Result 7.2).

Theorem 10.14 (Proposition C: self-consistency of a diffuse hole)**.** *A spectrally diffuse failure of the Weak Covering Conjecture satisfies the annealed ℓ^1 Fourier budget with slack ≈ 1.88; no Littlewood–Offord/Riesz-product counting argument of the kind used in [2] can exclude it. This* *branch — inaccessible to ℓ^1 methods by Theorem 10.14 — is exactly the branch identified as the core of $\beta = 1$ (§9) and of the refuted L^2 condition (Theorem 10.4): a further articulation, within the same second-moment Fourier vocabulary as the L^2 condition (both decompose via the character sum* $\mathrm{Cov} \propto \sum_\xi S_1(\xi)S_2(\xi)^*$ *of §8), of the same missing ingredient — not a methodologically independent fourth or fifth route in its own right.*

We emphasize the correct framing, consistent with the fragilities listed above: this is a reduction of the spectrally *concentrated* branch to the Erdős–Lagarias–Furstenberg family, together with a proof that the spectrally *diffuse* branch is inaccessible to ℓ^1 Littlewood–Offord methods — not a reduction of the Weak Covering Conjecture to the classical Z-number problem, which Theorem 10.14 and the discussion above show would be imprecise.

11 Contextualizing the barrier in the wider literature

A directed literature search, going beyond the Collatz-specific literature into the general theory of Fourier decay for self-similar measures, situates the barrier precisely and rules out several plausible-looking shortcuts, each for a specific, verified reason.

11.1 The Syracuse measure is a rigid 3-adic self-similar measure

Unrolling the renewal recursion of (1) gives, in the limit,

$$X = \sum_{k\geq 1} 3^{k-1}\, 2^{-S_k}, \qquad S_k := a_1 + \cdots + a_k, \tag{10}$$

the fixed point of the countable affine iterated function system $\phi_a(x) = 2^{-a}(1+3x)$ on $\mathbb{Z}_3$, weighted by $\Pr(a) = 2^{-a}$, every map sharing the *same* ultrametric contraction ratio $|3\cdot 2^{-a}|_3 = 1/3$. This is a genuine (if non-standard, countable, overlapping) self-similar structure, so the question of whether the general theory of Fourier decay for self-similar measures [25, 26, 27] transfers is a fair one to ask, and we checked it explicitly.

Proposition 11.1 (Rigidity of $\mathbb{Z}_3^\times$)**.** *Every infinite closed subgroup of $(\mathbb{Z}_3, +)$ has finite index (it equals $3^k\mathbb{Z}_3$ for some $k \geq 0$); consequently, since $\mathbb{Z}_3^\times \cong \mathbb{Z}/2 \times \mathbb{Z}_3$, every infinite closed subgroup of $\mathbb{Z}_3^\times$ has finite index. Since 2 is a topological generator of $\mathbb{Z}_3^\times$ (a primitive root modulo 3^n for every n), the unit phases 2^{-a} spread maximally, the opposite of the "thin multiplicative orbit" phenomenon (a Pisot-type obstruction to the Rajchman property) that drives non-uniform Fourier decay on the real line.*

Theorem 11.2 (The general theory does not transfer)**.** *(a) The pushforward mechanism of Baker–Banaji [25] (Archimedean stationary phase, requiring a C^2 map with $f'' \neq 0$ off a null set) would give, at best,* qualitative *Rajchman decay for the analogue considered here — strictly weaker than the super-polynomial decay already guaranteed by Proposition 1.1.*

(b) Brémont's Pisot-type criterion for non-Rajchman self-similar measures, transported to $\mathbb{Z}_3$, trivializes by Proposition 11.1: there is no room for a "thin" 3-adic multiplicative orbit of the kind the criterion requires.

(c) The renewal-theoretic technique of Li–Sahlsten [27], which needs two distinct *contraction ratios with a Diophantine relation between their logarithms, also degenerates: every branch of (10) has the* same *3-adic contraction ratio 1/3, so the relevant "scale walk" is deterministic ($S_n = n \log 3$, a pure lattice walk with no overshoot fluctuation), not the non-lattice random walk the technique is built to exploit.*

We stress that Baker–Banaji's general phenomenon — Rajchman measures with no uniform decay rate are generic, not pathological, in the class of self-similar/self-conformal measures — remains a fair piece of context: it shows that the kind of failure exhibited by Theorem 10.14 has real structural precedent in the reference literature, legitimizing the framing that a proof of $\beta = 1$/the Weak Covering Conjecture genuinely needs the specific arithmetic input of this problem, not generic smoothness. It should not, however, be over-interpreted as evidence *for* the diffuse scenario itself: Baker–Banaji's construction requires fine-tuning of a continuous parameter (a Liouville-type construction, dense in the family but exceptional in measure; typical parameters give polynomial decay [28, 29]), while the Syracuse measure is a rigid arithmetic object with no tunable parameter, and Proposition 11.1 shows the enabling mechanism (fine Diophantine approximation) simply does not exist on the 3-adic side — which cuts against the diffuse scenario as much as for it. We cite Theorem 11.2 with this disanalogy stated explicitly.

11.2 A fifth independent vocabulary, empirically tested

Very recent independent work [31] reduces the Collatz conjecture, by a purely combinatorial route with no Fourier analysis whatsoever, to a residue-balance condition along a sparse subsequence of the true (not ensemble) orbit: writing $X_t := \mathbb{1}[n_t \equiv 1 \bmod 4]$ for the compressed odd-to-odd orbit and identifying "burst-ending times" (the last index of a maximal run of $X_t = 1$), that work proves a Map Balance Theorem (the map-level bias between gap-length classes is exactly ± 1, for every $K \geq 5$) and reduces the conjecture, in the dominant residue class, to a claim that the fraction of burst-endings landing in one of two residues mod 32 stays bounded near $1/2$ along the orbit.

We tested this condition directly on real orbits. An ensemble of many moderate orbits (2–5×10^4 roots, 20–50 bits) shows the aggregate deviation plateauing at 1.2–1.9%, which is on its own consistent with the conjecture (only boundedness is claimed, not convergence to zero); a more faithful test — a single very long orbit (a random 16000-bit start, 38395 compressed steps, 4761 relevant burst-endings) — shows the deviation tracking the $1/\sqrt{i}$ statistical-noise curve closely, with no systematic trend, ending at a deviation of 0.00053. Five further orbits from 500 to 8000 bits show the same pattern. We interpret the ensemble plateau as an artifact of aggregating many short, finite, similarly-phased orbits, and the single-long-orbit result as qualified empirical support for the conjecture of [31], on the scale tested.

This gives a fifth, technically unrelated, vocabulary for the same missing ingredient: equidistribution/balance of residues along the true orbit, appearing independently via the smoothing-transform/ endogeny route (§4), the entropy count shared by the Weak Covering Conjecture and $\beta = 1$ (§7, §9 — proved to be the same algebraic object by Proposition 9.1, hence counted once), Wirsching's 2003 functional equation (§9), the second-moment Fourier analysis shared by the L^2 condition and the spectral dichotomy of Proposition C (§10, likewise counted once), and now a purely combinatorial reformulation.

12 Discussion: why a precisely characterized barrier is a genuine contribution

A natural objection to the material of §§4–11 is that neither candidate lemma of §10 produced a proof, so the paper might seem to document an absence rather than a result. We think this objection, while natural, does not survive scrutiny of what was actually produced.

First, the outcome of executing both candidate lemmas was not silence: it was two independent, rigorous negative/structural results, each of which pins down *precisely* why a natural approach fails

— Proposition 10.2's exact closed-form persistence of coarse correlation; Theorem 10.4's direct computational refutation of a specific sufficient condition; Theorem 10.13's exact identity quantifying, to a specific constant factor (≈ 1.88), why a whole family of Fourier-analytic techniques cannot close the gap. Precisely characterizing an obstruction — showing not merely that an approach fails, but *exactly* why, with a quantified margin — is itself a recognized and citable form of mathematical contribution, independent of whether the underlying conjecture is eventually resolved.

Second, these results do not stand alone: the same underlying arithmetic statement — some form of equidistribution or near-independence of 3-adic digits along the true, deterministic integer dynamics, not merely along an idealized i.i.d. model — is the crux reached by five methodologically independent vocabularies: (1) the smoothing-transform/endogeny route (§4); (2) the entropy count shared by Wirsching's Weak Covering Conjecture and Tao's $\beta = 1$, which Proposition 9.1 shows to be the *same algebraic object*; (3) Wirsching's 2003 real functional equation and its Fabius density (§9); (4) the second-moment Fourier analysis underlying both the L^2 condition and the spectral dichotomy of Proposition C (§10), which share the common character decomposition $\mathrm{Cov} \propto \sum_{\xi} S_1(\xi) S_2(\xi)^*$; and (5) the purely combinatorial residue-balance reformulation of [31]. We count five rather than seven deliberately: two of the pairwise linkages are not analogies but *proved identities* — $\beta = 1 \cong$ WCC (Proposition 9.1) and the common second-moment origin of the L^2 and spectral-dichotomy branches — and we count each such pair once. That two vocabularies can be *proved* to be one object is itself stronger evidence that the obstruction is a single structural feature of the problem than a loose analogy would be. A barrier visible from only one angle is a candidate artifact of that technique; a barrier reached by five independent vocabularies (branching-process fixed points, integer covering/entropy counting, a real functional equation, second-moment character sums, and elementary residue counting), two of whose pairwise coincidences are theorems, is much better evidence that the obstruction is a structural feature of the problem itself.

Third, this theoretical package sits alongside a result (Empirical Result 6.1) whose validity does not depend on any of the above framing at all: a direct, falsifiable, statistically calibrated empirical measurement resolving a long-standing dispute in the literature. We consider this the single most framing-independent contribution of the paper, and recommend it be weighted accordingly by a reader assessing the paper's overall reliability.

We therefore present this paper's contribution not as an attempted proof of any part of the Collatz conjecture, nor as a report of failure, but as a precise map of where a specific, natural, technically rich line of attack — Tao's Syracuse-measure program, generalized to $qx + 1$ — runs into a wall, together with an exact quantification of the size and shape of that wall from five independent angles, and one hard, unconditional empirical result obtained along the way.

13 Conclusion and open directions

We summarize the concrete open problems this paper leaves in the cleanest form we have been able to give them.

- (O1) **Near-independence of fresh 3-adic digits across sibling subtrees** (Theorem 4.3, Remark 4.4): the arithmetic statement whose idealized (i.i.d.) analogue is provable via smoothing-transform classification and Choquet–Deny rigidity.
- (O2) **Wirsching's Weak Covering Conjecture** (Conjecture 7.1), equivalently (Proposition 9.1) Tao's $\beta = 1$ (§9), open since 1998 and 2020 respectively.
- (O3) **Wirsching's Conjectures 1 and 2** (§9), above the numerically-tested Conjecture 3, of the same central-limit-window difficulty.

(O4) **Exponential, ξ-uniform Fourier decay for the Syracuse measure at depth $\ell \asymp D$** (§8, Regime 3), equivalent to recognized open problems in effective $\times 2, \times 3$ rigidity and Fourier decay of self-similar 3-adic measures.

(O5) **Exclusion of spectrally diffuse Weak Covering Conjecture failures** (Theorem 10.14), shown here to be inaccessible to ℓ^1 Littlewood–Offord methods, leaving genuinely new technique as the only route.

(O6) **Chang's residue-balance conjecture** [31] (§11), supported empirically here on a single very long orbit, unproved.

(O7) **The tail index of the martingale scale factor** (Conjecture 3.10), for every $q \geq 5$: unlike the $q = 3$ case, this still has no rigorous derivation (the relevant root is always frozen, Proposition 3.5). Empirical support, however, has substantially strengthened since the original under-powered measurement: a $20\times$ larger sample (10^5 roots, the pre-registered comparison called for above) produced, for the first time in this investigation, a clean GPD threshold-stability plateau at the predicted value across every threshold level tested, a tightly concentrated bias-corrected Hill estimate consistent with it across all headroom levels, and a Vuong test against a truncated lognormal alternative that no longer favors the alternative. Two calibration checks (against an exact synthetic Pareto sample of the predicted index, and against a deliberately mis-specified normalization exponent) found no artifact behind this shift. We regard the conjecture as now strongly supported for $q = 5$, though not proved: the Vuong test yields non-rejection rather than confirmation, and the underlying martingale is provably still pre-asymptotic at the headroom reached (its median continues to drift with headroom even as the estimated tail index does not). This was, honestly, the weakest-supported claim in the paper; it no longer is, but it remains the only one resting on statistical evidence rather than a closed derivation.

(O8) **The density transition on the genuine arithmetic tree** (Conjecture 3.8): established here only for the matching i.i.d. branching-random-walk model (Theorem 3.6, and, conditionally, Proposition 3.7); for $q = 3$ this conjecture implies, and strictly refines, the Growth Exponent Conjecture of Kontorovich–Lagarias and Applegate–Lagarias, open since 1995, of which the best unconditional partial result remains the one-sided bound of Krasikov–Lagarias [16].

Any progress on any one of (O1)–(O8) would very likely transfer, given the structural identities established in this paper, to progress on several of the others simultaneously. We regard this convergence as the paper's central finding.

Code and data availability

Every computational result reported in this paper — the pressure and freezing computations of §3, the residual-coupling control experiment of §5, the Kontorovich–Lagarias vs. Volkov gate of §6, the Weak Covering Conjecture test of §7, the Wirsching (2003) Conjecture 3 test and the L^2/Jensen computations of §9–§10, and the Chang residue-balance test of §11 — has an accompanying, independently runnable script, organized by section, at `https://github.com/faculdade/collatz-endogeny`. Each subfolder of that repository has its own README stating what it verifies, how to run it, and the expected output.

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A Numerical validation methodology

We record, for completeness, the validation discipline applied to each computational result reported above, since an unvalidated recursive computation is a known source of silent error.

- *Pressure equation roots* (§3): solved by bisection independently of the closed-form derivation; cross-checked against empirical counting slopes on real reverse trees for $q = 5, 7$.
- *Weak Covering Conjecture DP* (§7): validated against an independently computed reference table for $\ell = 1, \ldots, 7$ and a hand-checked point ($\ell = 2, j = 2$).
- *K_ℓ recursion* (Theorem 10.4): the only initially available check was the hard-coded base case ($K_1 = 5/3$), which does not exercise the recursive machinery; we additionally validated the full distribution μ_ℓ, bin by bin, against direct Monte Carlo sampling of the primary definition (1) at $\ell = 3, 4$, within Monte Carlo noise.
- *Jensen-identity Monte Carlo* (Theorem 10.13): the identity is exact and independent of simulation; the Monte Carlo estimate (8×10^6 samples) serves only as a sanity check on the analytic value, and matches it to four significant figures.
- *Wirsching 2003 moment recursion* (Empirical Result 9.4): validated by an independent fixed-point check on an independent grid, agreeing to the grid's resolution (≈ 6 digits), and by confirming the functional equation (7.12) of [5] numerically up to $\ell = 200$ using an analytic (not numerically differenced) logarithmic derivative.
- *Chang residue-balance test* (§11): validated by confirming that no burst-ending event ever produced a residue outside the two predicted values ($9, 25 \bmod 32$) in either the ensemble or single-orbit experiments, matching the source paper's structural proposition exactly.

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